Planejamento de Capacidade
Técnicas e Ferramentas

Rodrigo Campos
camposr@gmail.com - @xinu

Agenda

- Planejamento de Capacidade
- Métricas
- Filas
- Modelos
- Ferramentas

Planejamento de Capacidade

- Todos querem fazer
- Muitos dizem que estão fazendo
- Você sempre exagera ao dizer o quanto está fazendo
- A maioria das pessoas não está realmente fazendo (apesar de seu esforço)
- Todos parecem fazer mais do que você...

Planejamento de Capacidade

Uma história de sobrevivência...

Planejamento de Capacidade

QUANTOS?

Servidores precisamos?
Gigabytes de memória?
Petabytes de Storage?
GHz de Processamento?
Gbps de banda?
IOPS de leitura/escrita?

Por que eu deveria me preocupar com isso?

Por que eu deveria me preocupar com isso?

Em 2012...

Por que eu deveria me preocupar?

Capacity planning is not just about the future anymore.

Today, there is a serious need to squeeze more out of your current capital equipment.

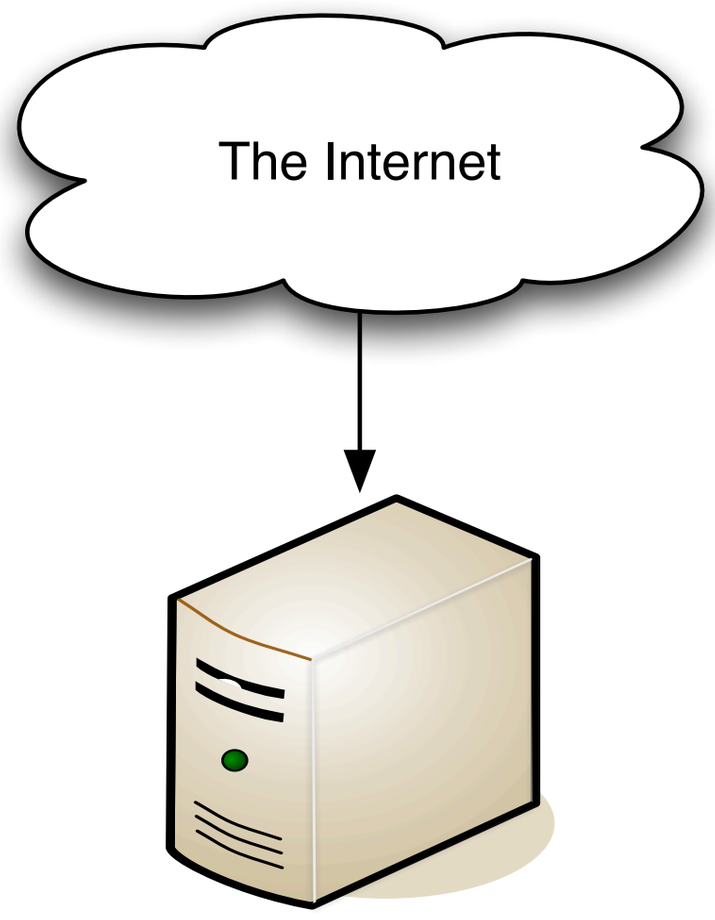
The Guerrilla Manual Online

http://www.perfdynamics.com/Manifesto/gcaprules.html

Por que eu deveria me preocupar?

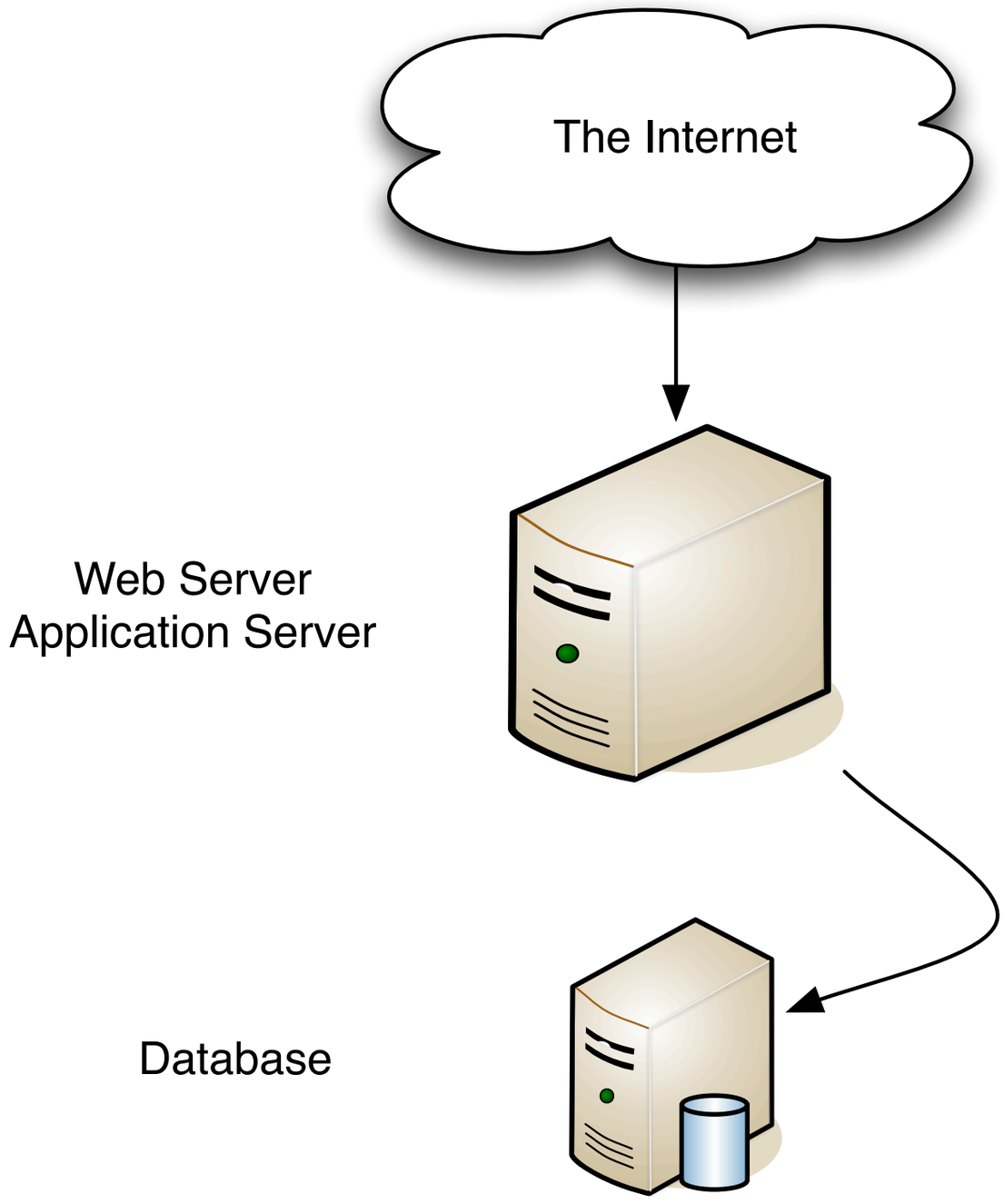
"Nossos sistemas são muito simples, não precisamos de planejamento de capacidade"

Começa assim...

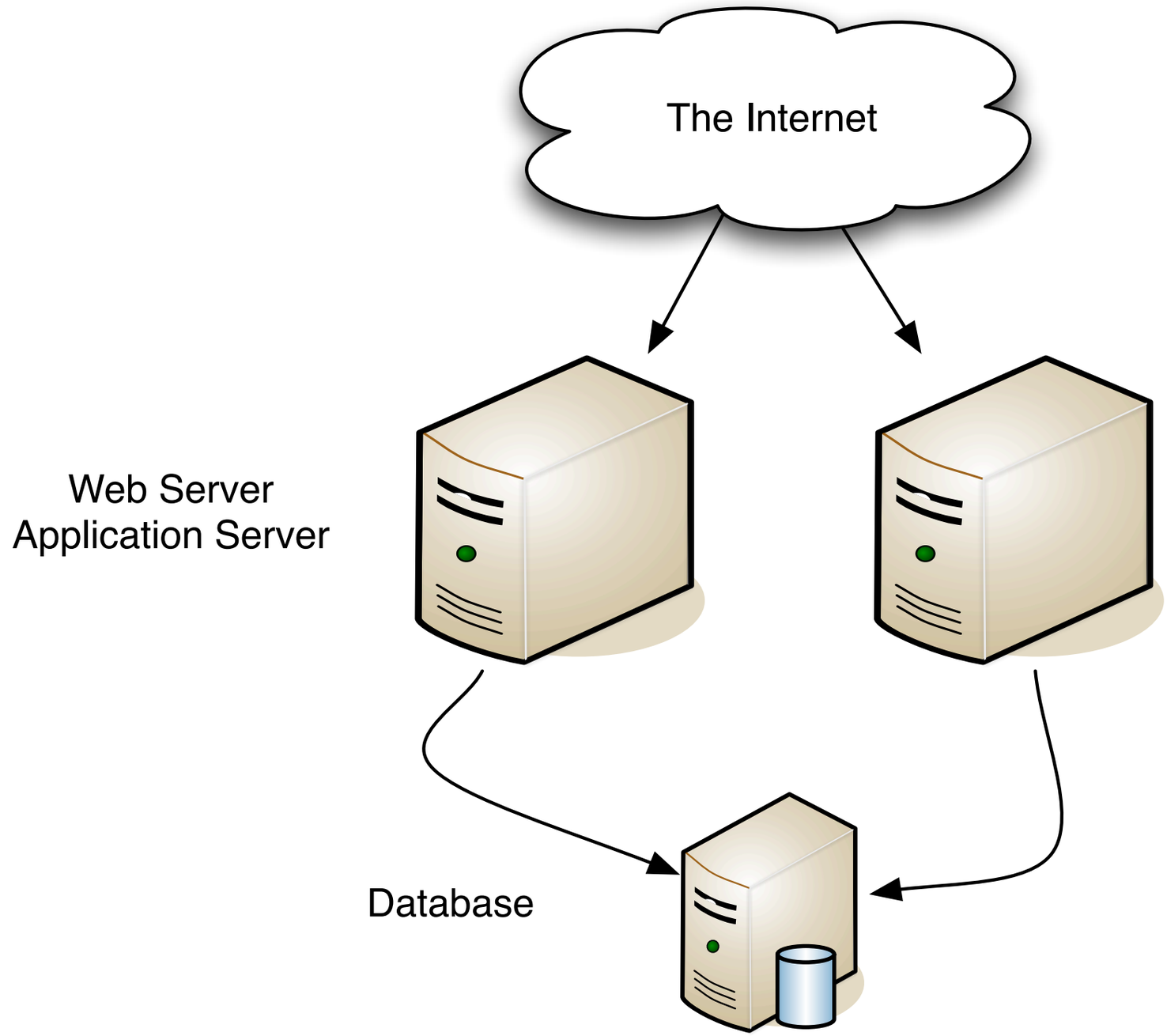


Web Server
Application Server
Database

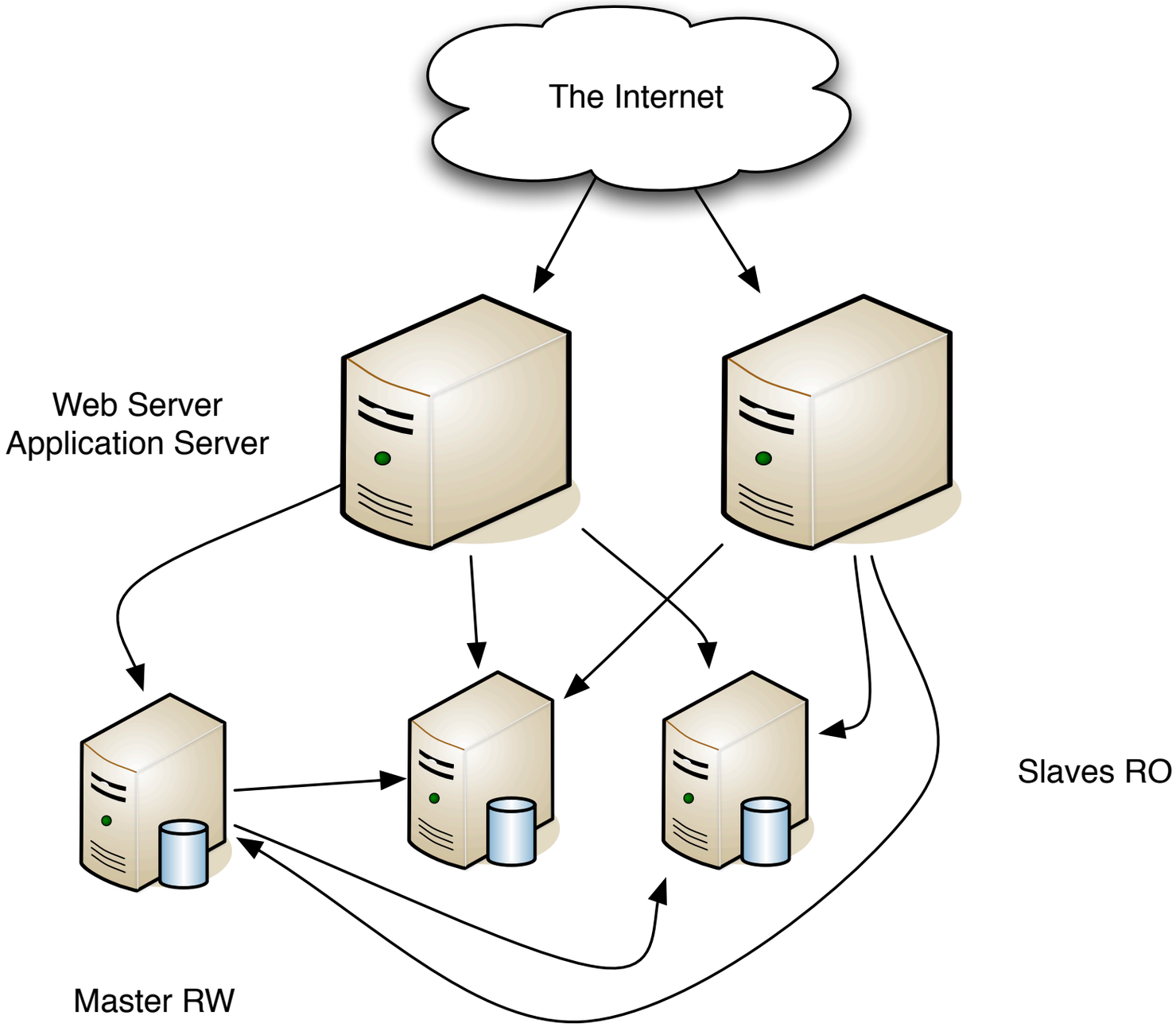
Depois assim...



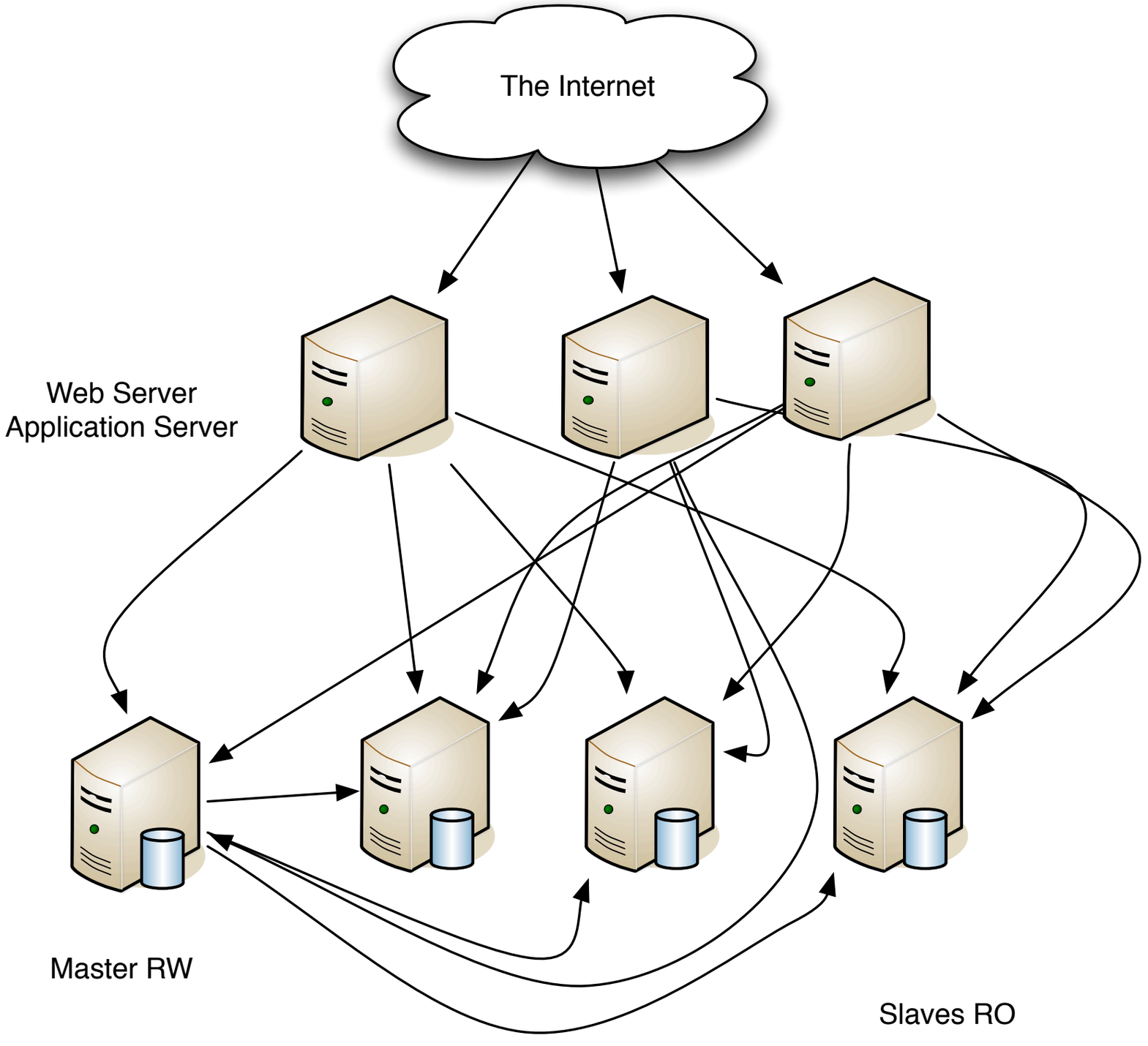
E assim...



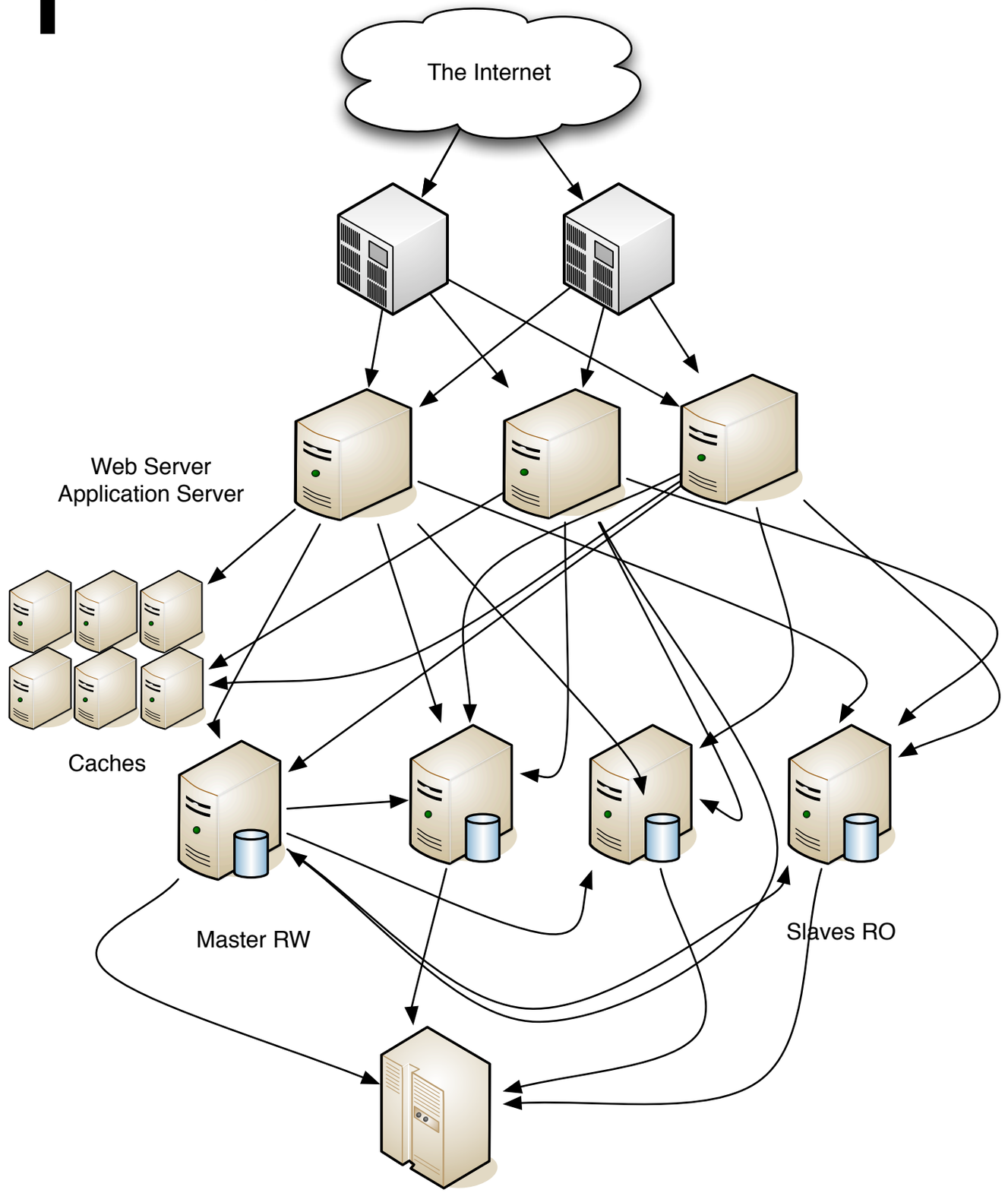
Um mês depois...



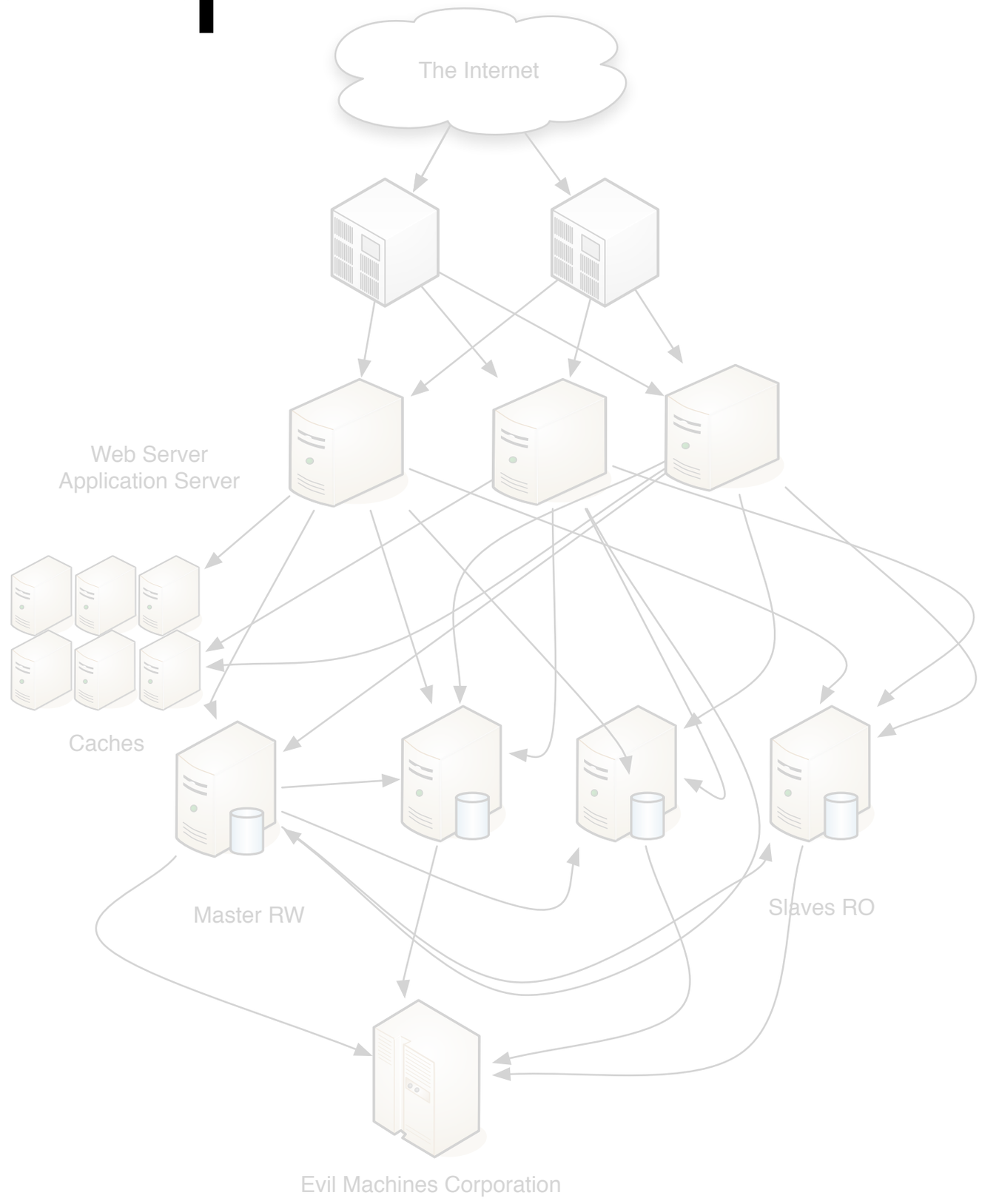
E ainda...


The Internet
Web Server
Application Server
Master RW
Slaves RO

Depois de 3 mêses...

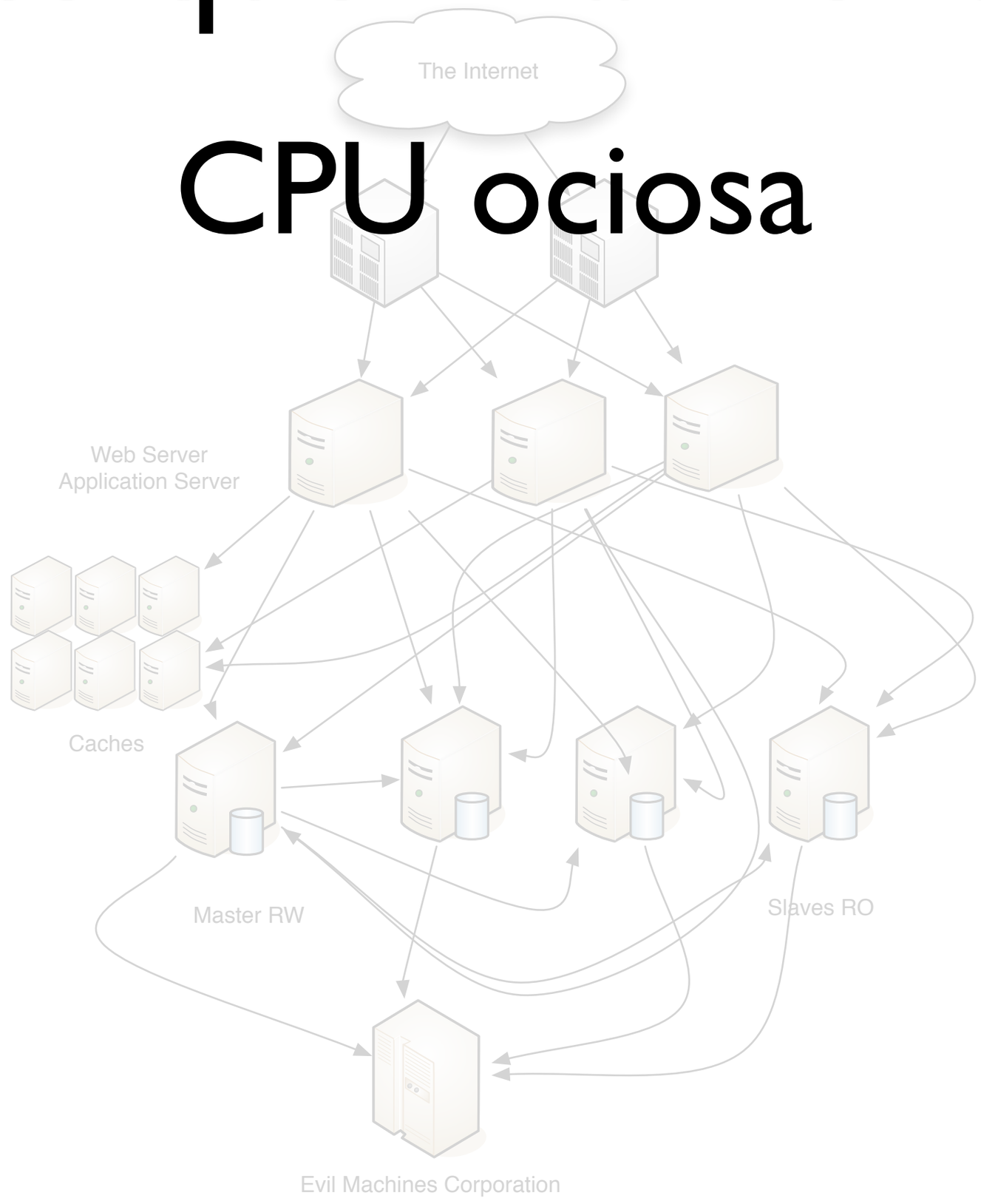


Até que finalmente!



Até que finalmente!

CPU ociosa



Até que finalmente!

CPU ociosa
Discos dormindo



Até que finalmente!

CPU ociosa

Discos dormindo

Rede subutilizada



Até que finalmente!

CPU ociosa

Discos dormindo

Rede subutilizada

Memória sobrando



Até que finalmente!

CPU ociosa

Discos dormindo

Rede subutilizada

Memória sobrando

... e os **USUÁRIOS**

reclamando!!!!

Como você explica isso?!?!?!



Por que eu deveria me preocupar?

"Agora nós estamos na nuvem!"

Por que eu deveria me preocupar?

- Bem vindo ao mundo do "Utility Computing"
- Onde você é cobrado **pelo uso!!!** (de novo)

Cloud Computing

"If computers of the kind I have advocated become the computers of the future, then computing may someday be organized as a public utility just as the telephone system is a public utility... The computer utility could become the basis of a new and important industry."

John McCarthy - 1961

Cloud Computing

"While all of this was going on, we began to hear stories out of the Pentagon about an approach to data processing and computing being fostered by my old acquaintance Grace Hopper. ***Large computers were going to be replaced by networks of small ones capable of communicating with one another and with common databases that might be on computers of any size.*** *(...) Also, any user at any node could access data on any database in the network with the same provisos." (circa 1970)*

My Adventures with Dwarfs - Russel C. McGee

Cloud Computing

1. Consumo não previsto de recursos
2. Máquinas virtuais "esquecidas"
3. Escolha de um modelo de custo errado
4. Custos de manutenção

4 Causes Of Cloud Bill Shock

Some companies shoot for reduced costs only to find unexpected expenses drive up the cloud computing bills. Do any of these problems lurk in your company?

By **Beth Stackpole** InformationWeek
May 29, 2012 09:05 AM

While the promise of the cloud is more cost-effective IT, the reality is that once companies move beyond simple deployment into a wholesale enterprise architecture shift, they often bump up against unanticipated expenses that threaten the cloud's principal value proposition.

It could be a development team that spins up hundreds of servers for load testing that they then forget to switch off, or a marketing team reserving dozens of new server instances to run a specific campaign when there are untapped enterprise resources they could otherwise deploy. Whatever the case, the 'aha moment' around the cloud's hidden costs typically presents itself along with a hefty bill that can leave some CFOs questioning the long-term viability of the cloud as a cost-effective platform.



Amazon's 7 Cloud Advantages: Hype Vs. Reality

(click image for larger view and for slideshow)

"It's the shocking bill problem--that's when the pain first gets raised," said Sharon Wagner, CEO of Cloudyn, a provider of an automated cost management solution for analyzing cloud spending.

It's a scenario many early cloud adopters are just now confronting as their cloud usage and infrastructure expands as, conversely, their direct visibility and control over usage patterns shrinks. Unlike days past, CIOs are unable to maximize cloud resources because they likely don't know what applications are running, where they are running, how much they are consuming, and what the cost implications are.

[Allocating costs across the enterprise has always posed problems, but cloud computing brings new complications. See Cloud's Tough Enemy: Chargeback Pushback.]

More Cloud Insights

Webcasts

- Creating an Agile, Flexible Cloud Computing Model
- Maximize ROI with Database Consolidation onto Private Clouds

More >>

White Papers

- e-Commerce Strategies for Business-to-Business (B2B) Sales and Marketing
- Cloud Computing Drives Break through Improvements in IT Service

http://www.informationweek.com/news/cloud-computing/software/240001065

Por que eu deveria me preocupar?

"Atualizar contadores de desempenho vão deixar o meu código mais lento"

Por que eu deveria me preocupar?

- Utilização média de CPU em um datacenter típico: 15%
- Se atualizar métricas são um problema, é um sinal de que você precisa delas
- Aqueles microssegundos irão salvar horas de manutenção

Por que eu deveria me preocupar?

"Métricas são requisitos não funcionais"

Por que eu deveria me preocupar?

	Distinct Queries/User	Query Refinement	Revenue/ User	Any Clicks	Satisfaction	Time to Click (increase in ms)
50ms	0	0	0	0	0	0
200ms	0	0	0	-0,30%	-0,40%	500
500ms	0	-0,60%	-1,20%	-1,00%	-0,90%	1200
1000ms	-0,70%	-0,90%	-2,80%	-1,90%	-1,60%	1900
2000ms	-1,80%	-2,10%	-4,30%	-4,40%	-3,80%	3100

The User and Business Impact of Server Delays, Additional Bytes, and HTTP Chunking in Web Search - Eric Schurman (Amazon), Jake Brutlag (Google)

http://velocityconf.com/velocity2009/public/schedule/detail/8523

Por que eu deveria me preocupar?

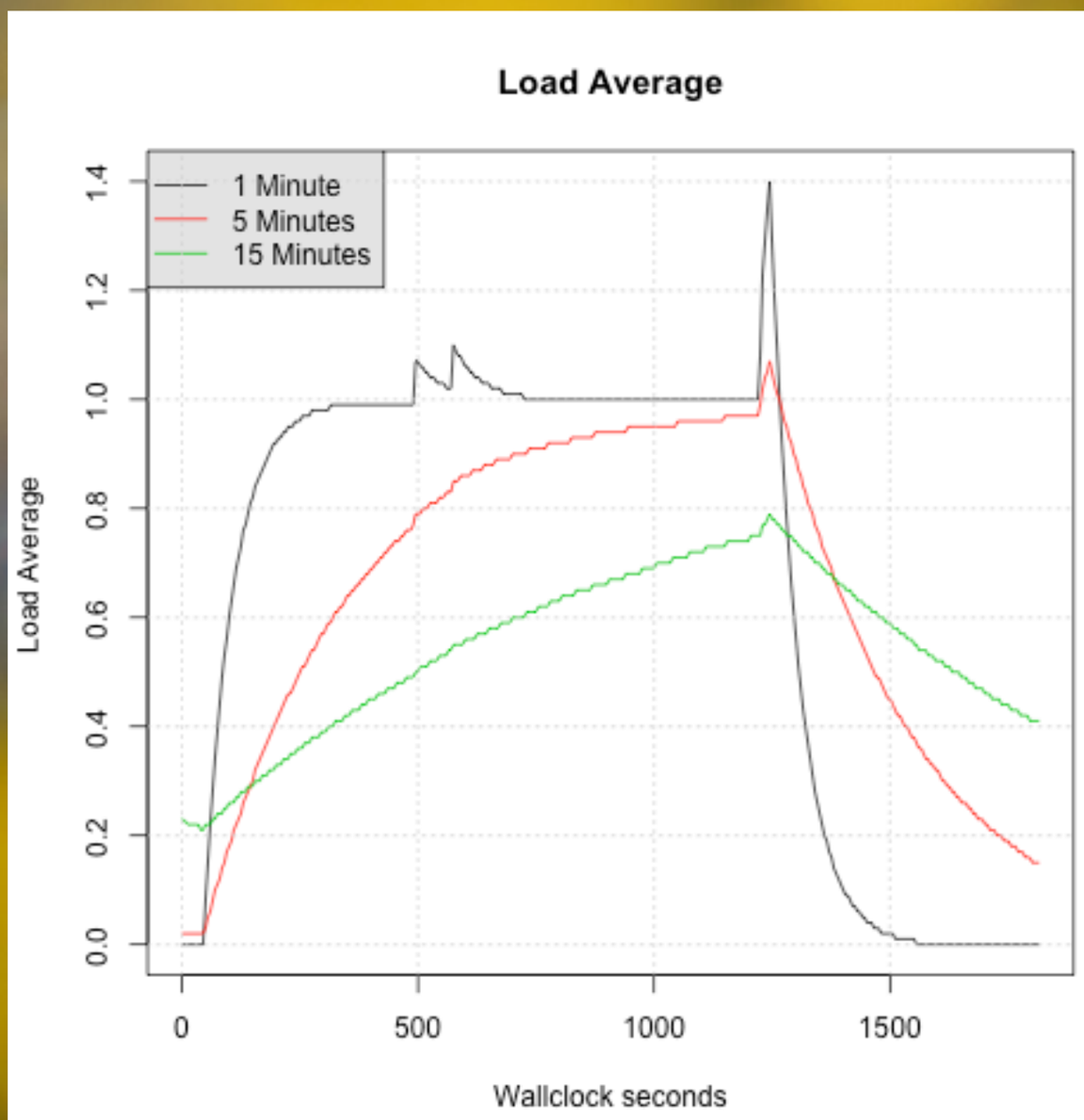
"Fast isn't a feature, fast is a Requirement"

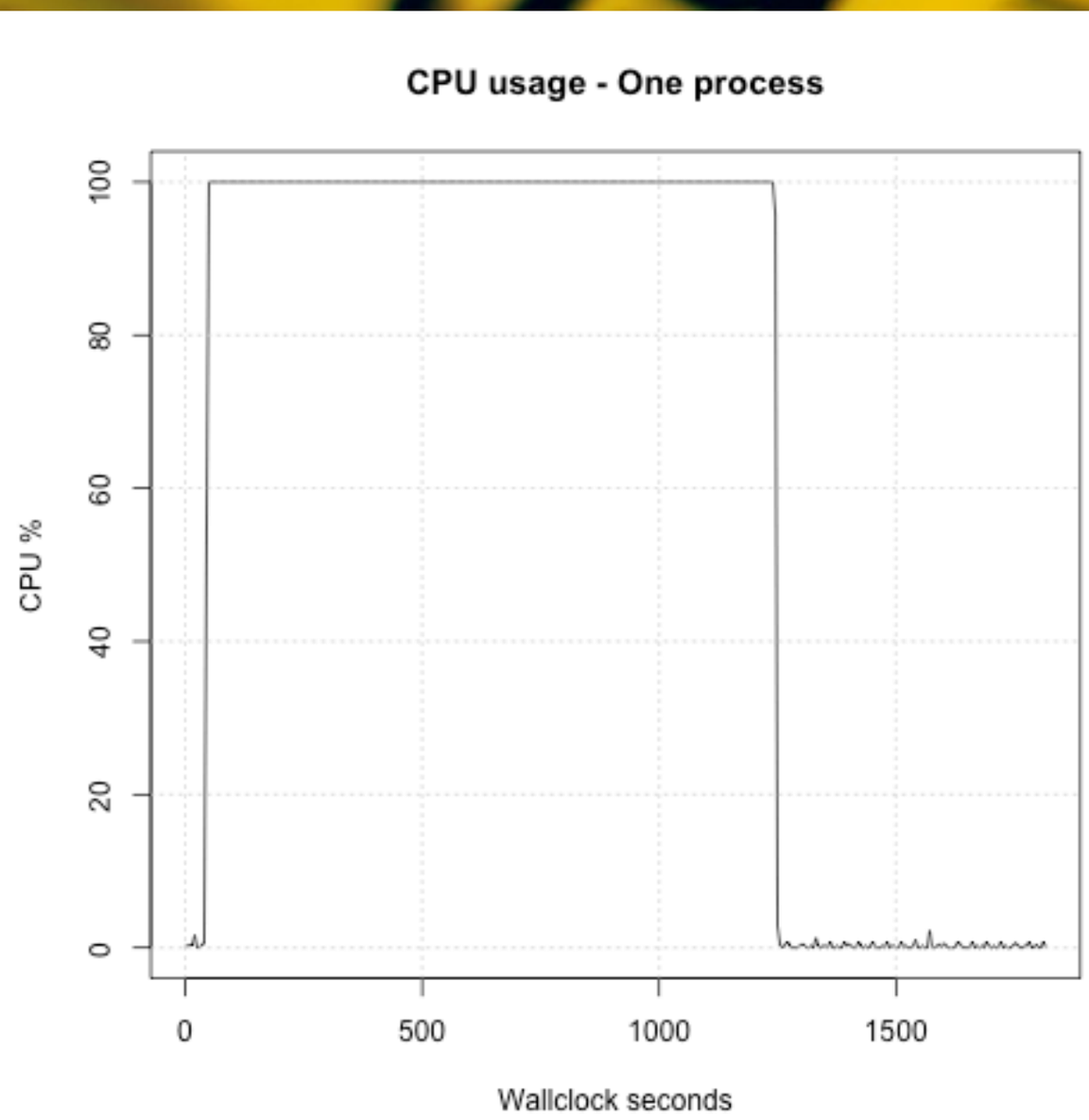
Jesse Robins - OPSCode

Métricas de Desempenho Típicas

- Load Average & Processor Queue Size
  - A métrica mais incompreendida do mundo?
- CPU
  - %user, %system, %iowait
- Dispositivos de IO
  - Bytes read/write & Blocks read/write
- Memória
  - Pages in/out & Swap in/out

Métricas de Desempenho Típicas





Séries Temporais

- Séries temporais servem para:
  - Troubleshooting
  - Previsões simples
  - Encontrar tendências (do sistema atual)
  - Identificar comportamento sazonal
- **Apenas isso não é planejamento de capacidade**
- **Não permite a criação de cenários hipotéticos**

Chamem os especialistas

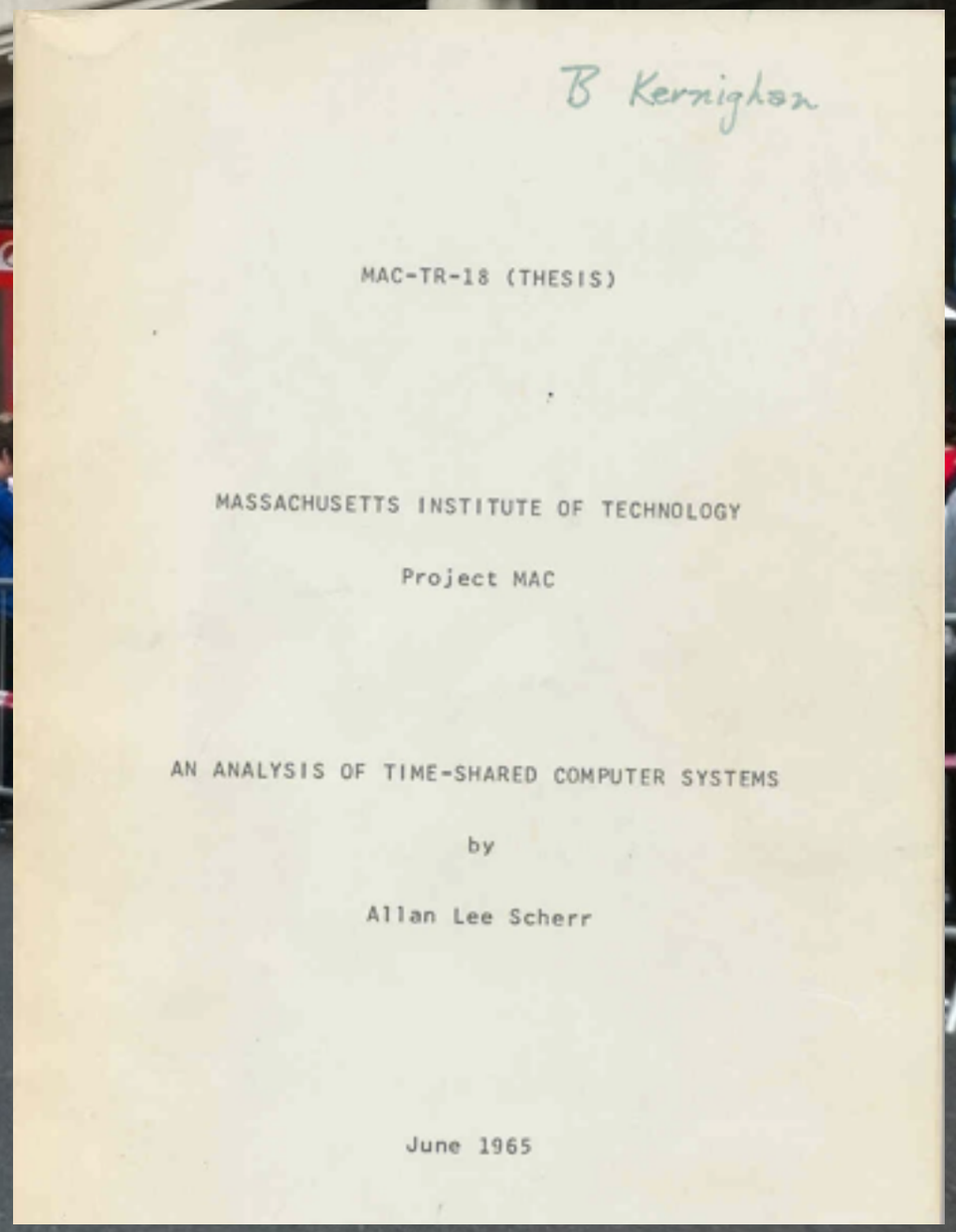
- Custo por MIPS
- IBM System/370 model 158-3 - 1.0 MIPS @ 1.0 MHz -1972
  - Preço Médio: $ 771,000*
    - Sem discos ou periféricos
  - $ 4,082,039 em 2011
- Necessidade de extrair toda a capacidade do sistemas

** Source: http://www-03.ibm.com/ibm/history/exhibits/mainframe/mainframe_PP3135.html*

Filas

A métrica de desempenho esquecida

- 1961 - CTSS foi demonstrado no MIT
- 1965 - Allan Scherr utilizou o problema do reparador para modelar um sistema de *timesharing* no Projeto MAC



B Kernighan

MAC-TR-18 (THESIS)

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Project MAC

AN ANALYSIS OF TIME-SHARED COMPUTER SYSTEMS

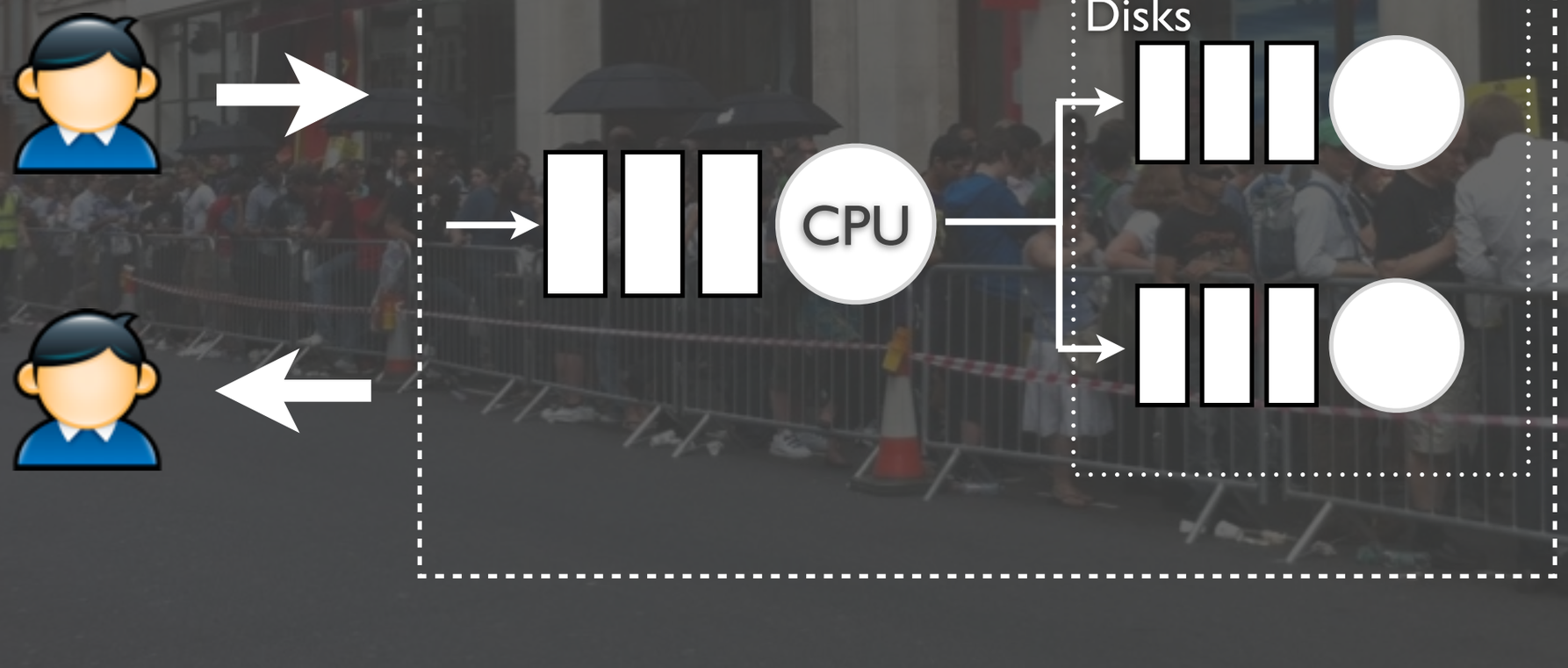
by

Allan Lee Scherr

June 1965

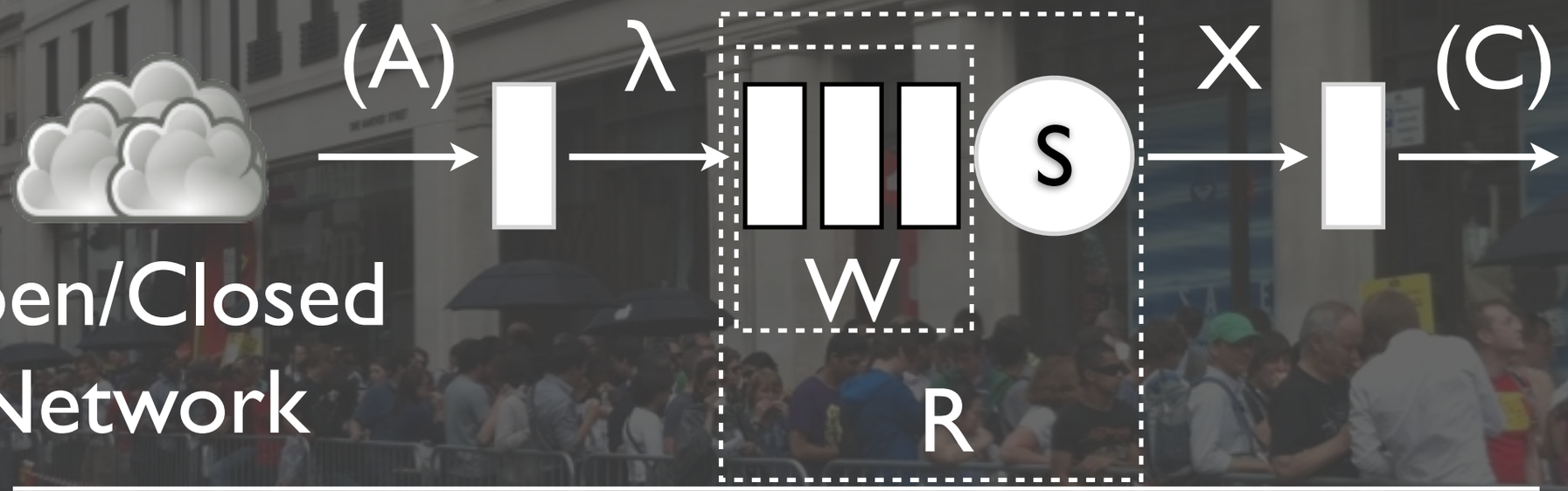
Filas

A métrica de desempenho esquecida



Filas

A métrica de desempenho esquecida



Open/Closed Network

A	Arrival Count
λ	Arrival Rate (A/T)
W	Time spent in Queue
R	Residence Time (W+S)
S	Service Time
X	System Throughput (C/T)
C	Completed tasks count

Arrival Rate (λ)

- Taxa de chegada de requisições
- Requisições por segundo/hora...

$$\lambda = \frac{A}{T}$$

- Não é a mesma coisa que *throughput*
  - Exceto em uma fila estável:
    - $A = C$ as $T \rightarrow \infty$
    - $\lambda = X$

Service Time (S)

- Tempo de Serviço
- Tempo gasto em processamento
- Tempo de resposta da aplicação
- Tempo de resposta de uma query
- Duração de uma operação de IO

Desempenho Mítico

- Só existe nas ciências biológicas ou em lendas
- Não acredite em tudo o que lê
- *"In God we trust, all others must bring data" - William Edwards Deming*

Como medir?

- Apache: %D in mod_log_config
- nginx: $request_time in HttpLogModule
- use Benchmark;
- tcprstat - http://goo.gl/0cbYx
- collectd - http://goo.gl/OXKG7
- metrics - http://goo.gl/gQFVM
- sysstat - http://goo.gl/2aLul

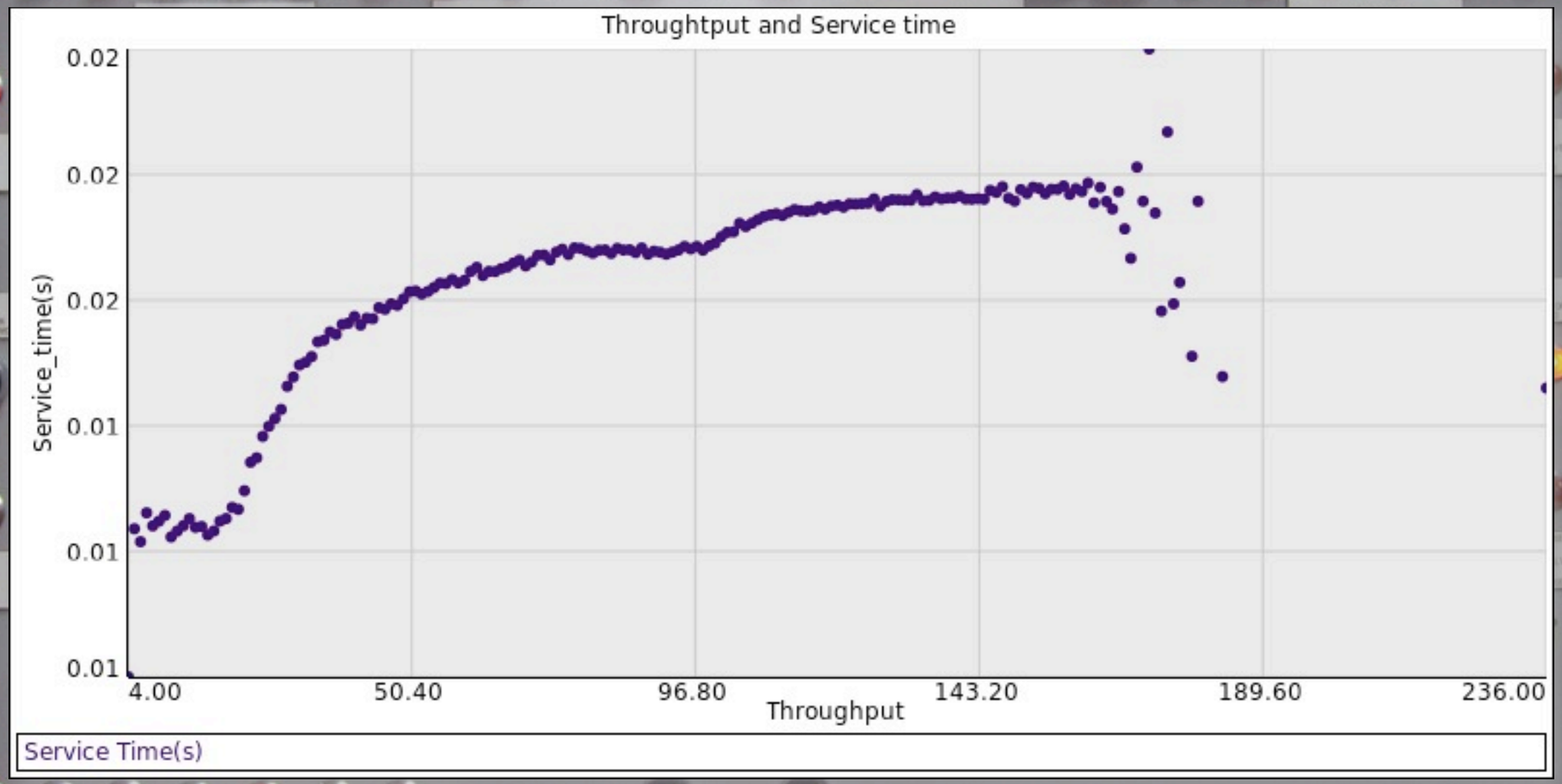
Como medir?

```
my ($date,$svctime) = (m/\[(\S+).+?\s(\d+)$/);
$arrivalRate{$date}++;
$serviceTimeAcc{$date} += $svctime;
```

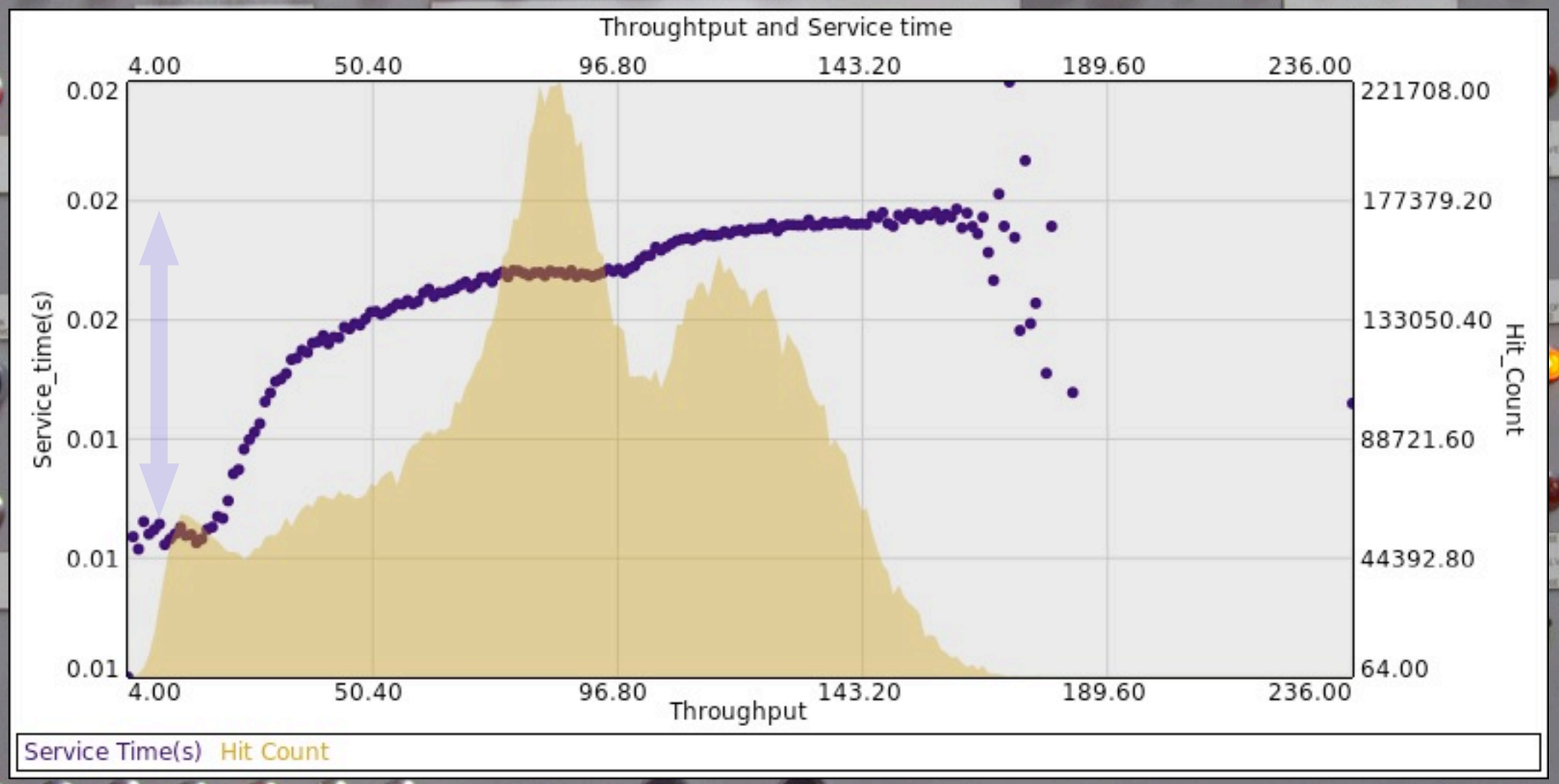
```
[02/Jul/2010:14:00:18... 1863
```

Tempo para atender a requisição em µsegundos.

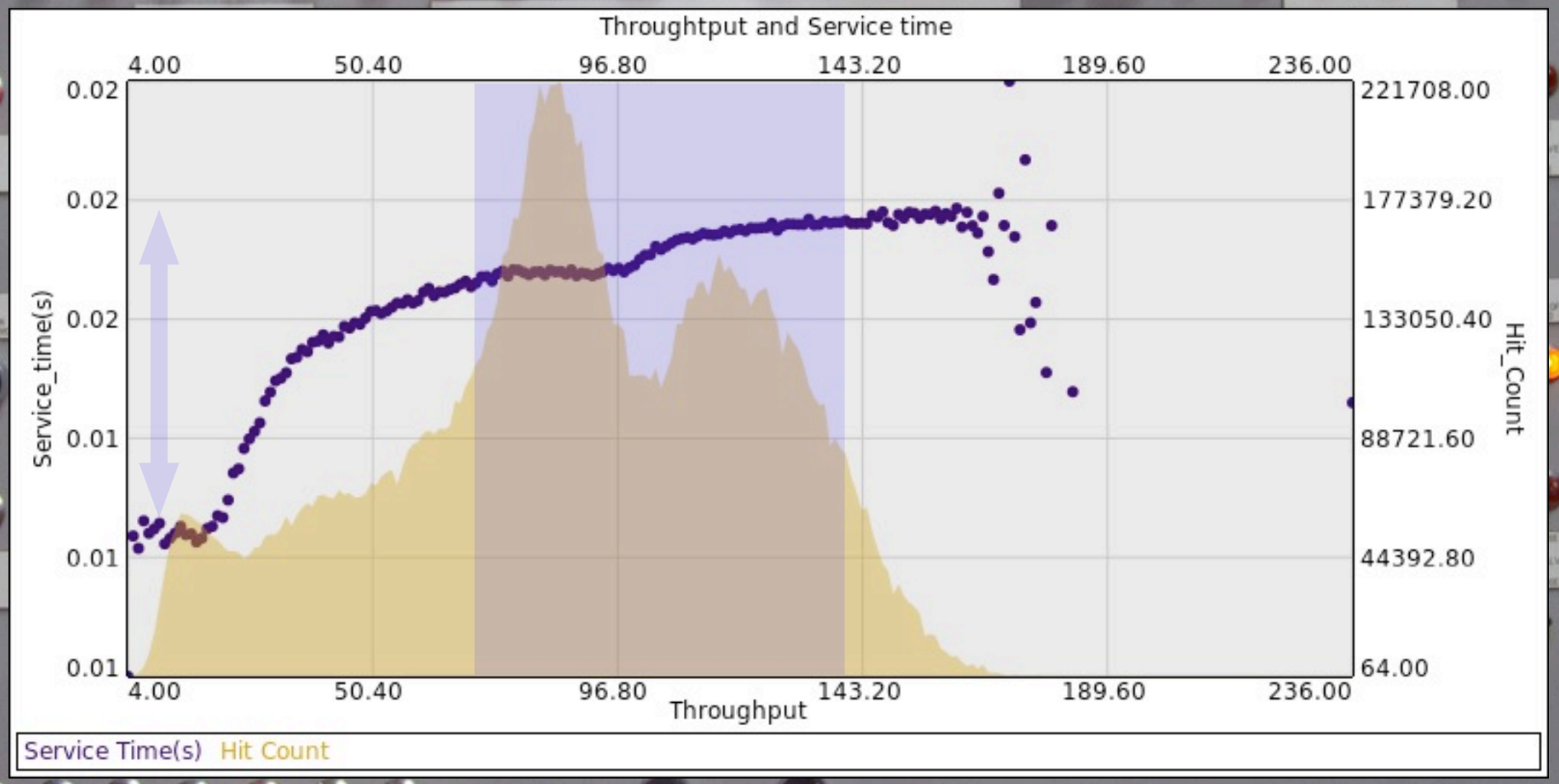
Visualizando o desempenho



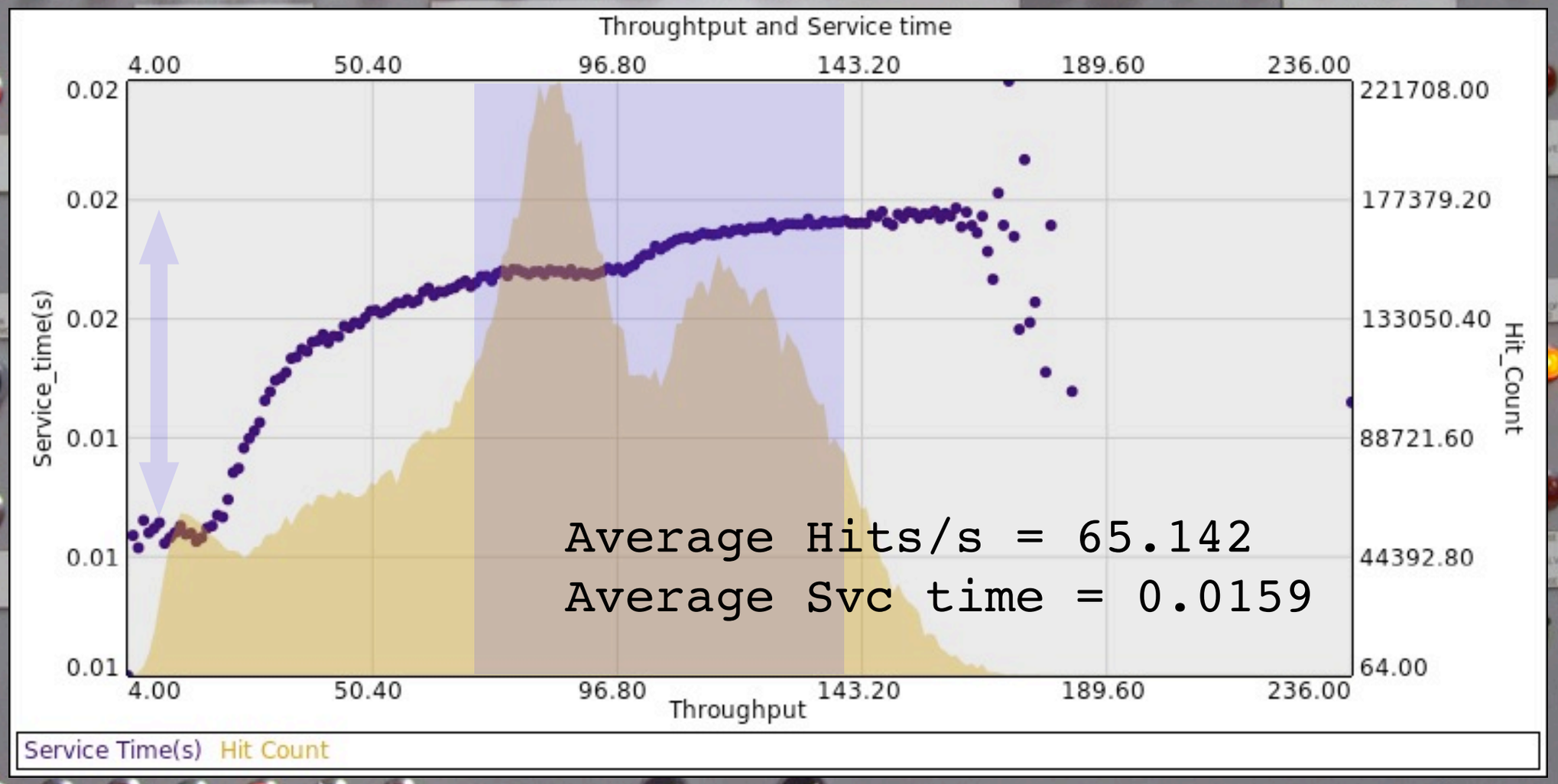
Visualizando o desempenho



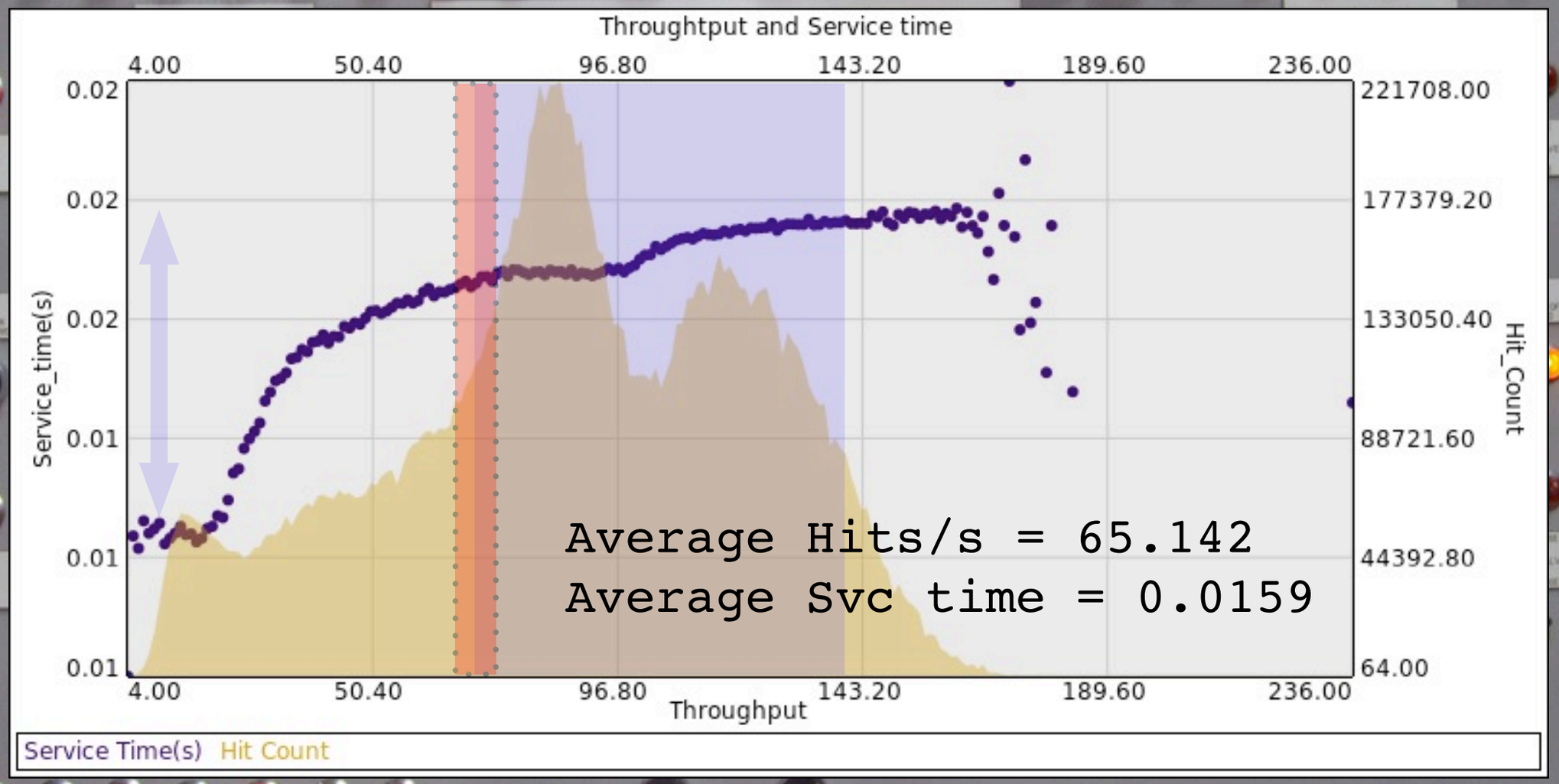
Visualizando o desempenho



Visualizando o desempenho



Visualizando o desempenho



O que devemos observar?

- *Stretch factor*
- Métodos e operações envolvidas
- Origem da requisição
- Tráfego
- Latência

Modelagem

Prediction is very difficult, especially if it's about the future.

Niels Bohr

Capacity planning is about setting expectations. Even wrong expectations are better than no expectations!

Neil J. Gunther - The Guerrilla Manifesto

http://goo.gl/lZKWH

Modelagem

- Um modelo é uma abstração de um sistema complexo
- Um modelo permite a observação de comportamentos que dificilmente seriam replicados

Métodos de Modelagem

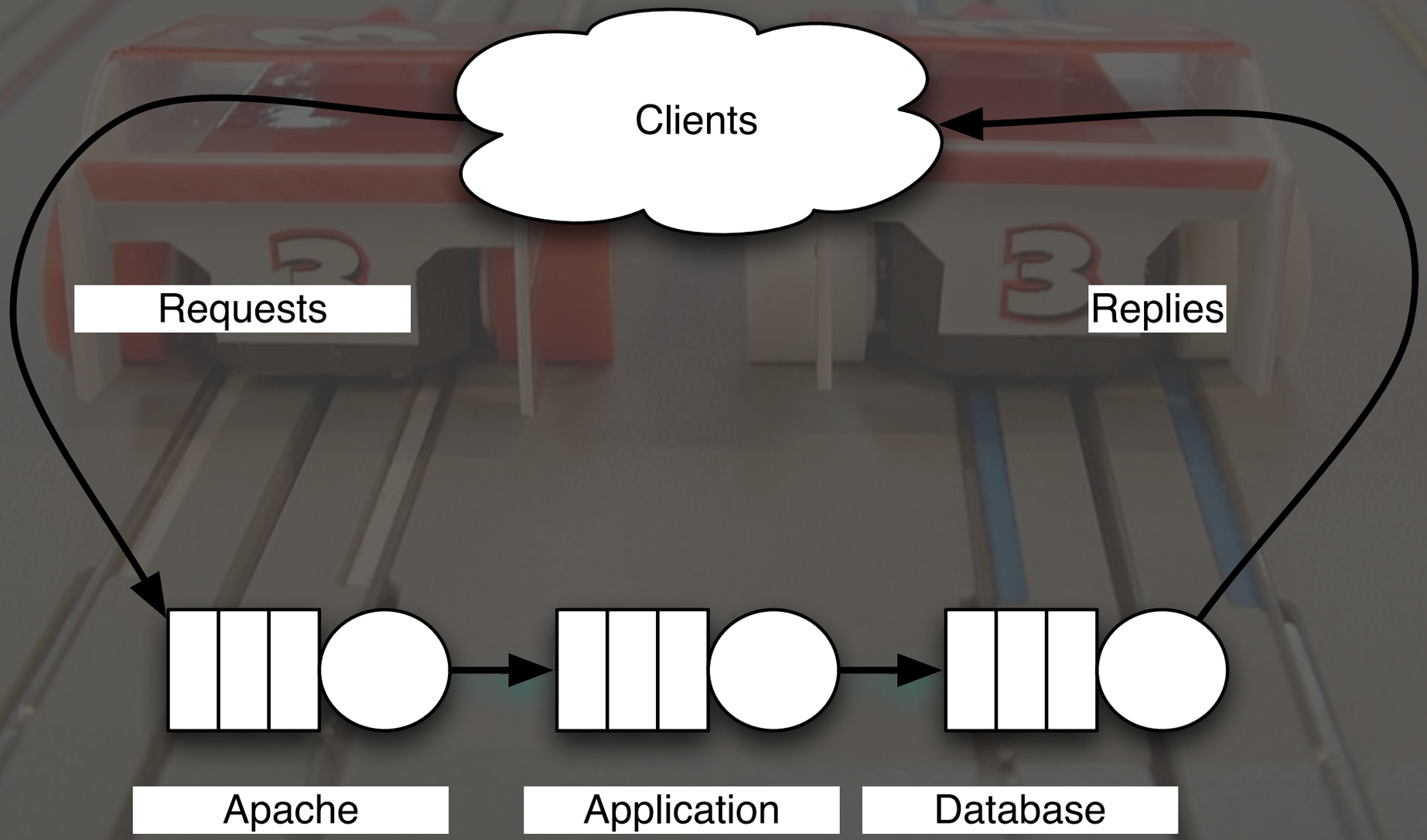
- Estatística / Trending / Forecasting
  - Vantagens:
    - Fácil compreensão
    - Ferramentas amplamente disponíveis
  - Desvantagens:
    - Não permitem a criação de cenários hipotéticos
    - Dificilmente demonstram contenções e limites

Métodos de Modelagem

- Análise de filas
  - Vantagens:
    - Permitem previsões mesmo quando os dados de produção são limitados
    - Permitem a criação de cenários hipotéticos
  - Desvantagens:
    - Não são intuitivos
    - A matemática pode ser complexa

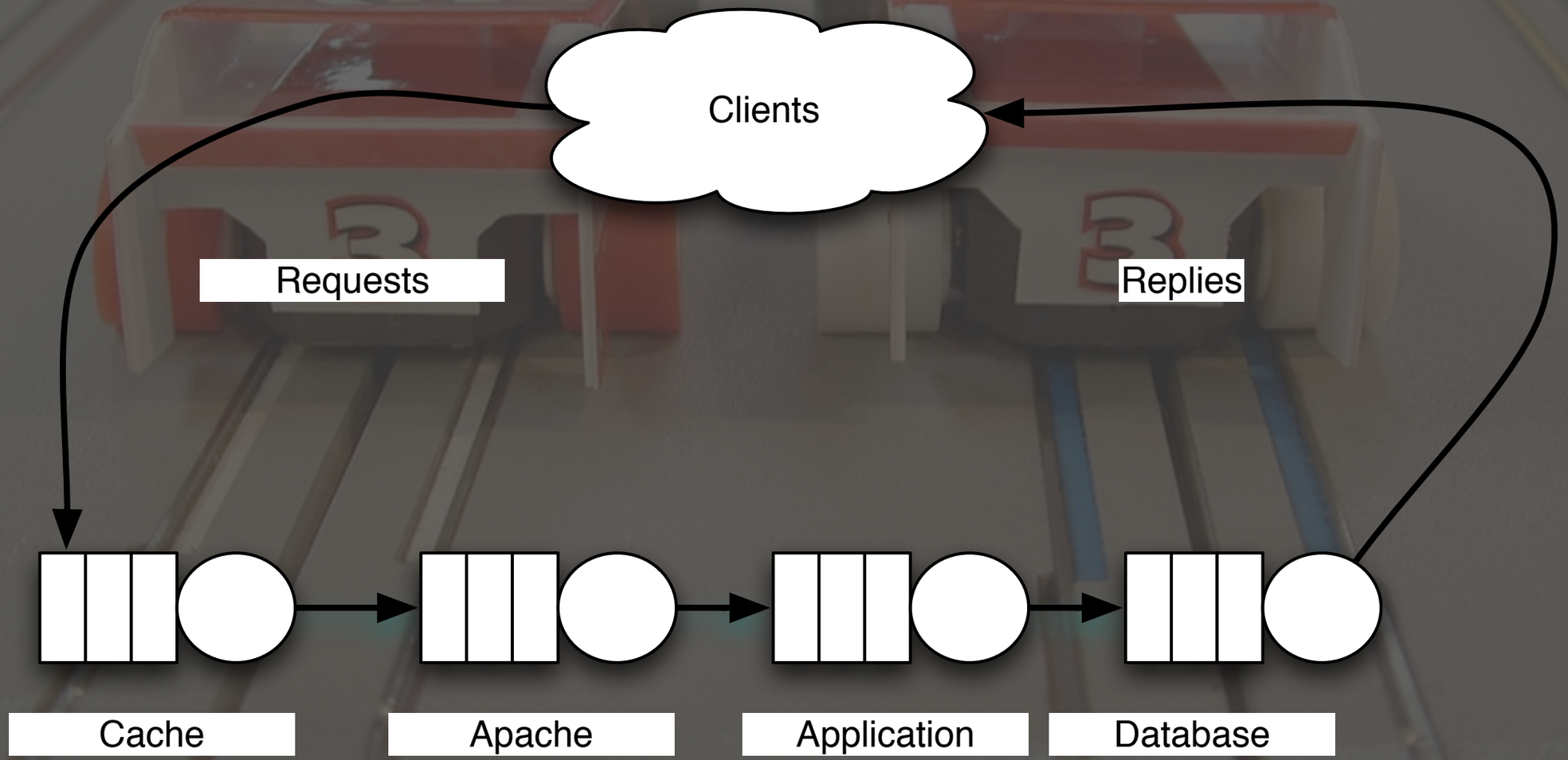
Filas como Modelos

Típica instalação LAMP



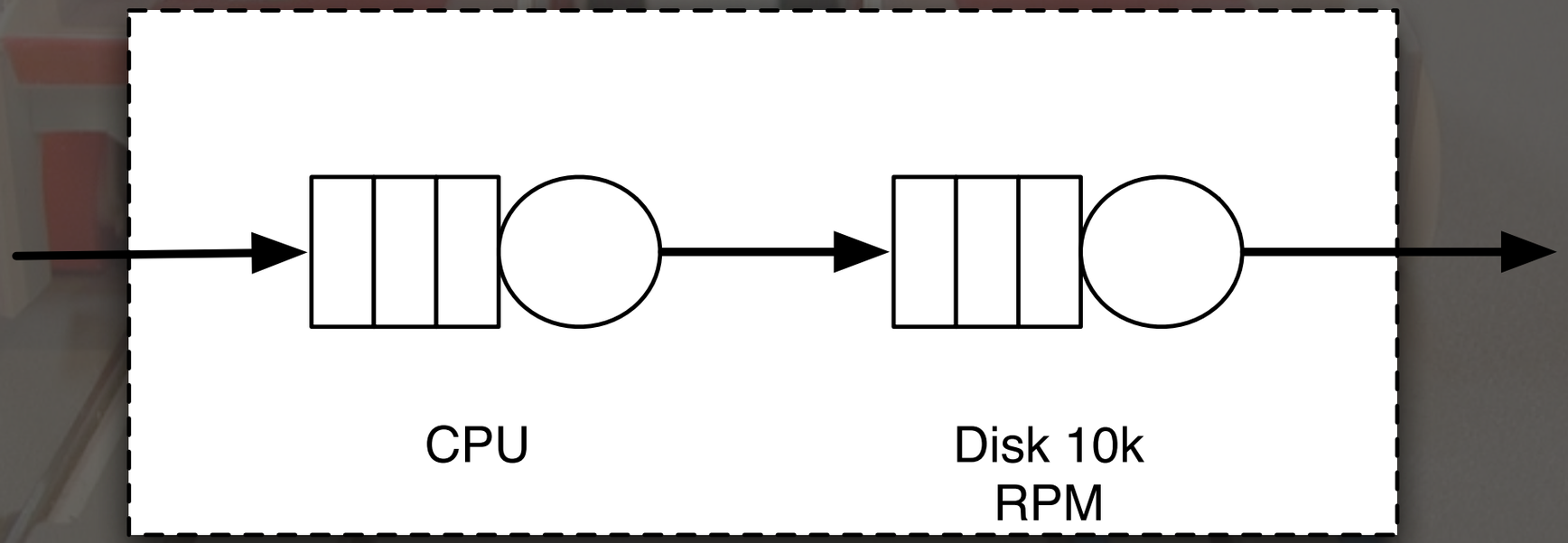
Filas como Modelos

E se?



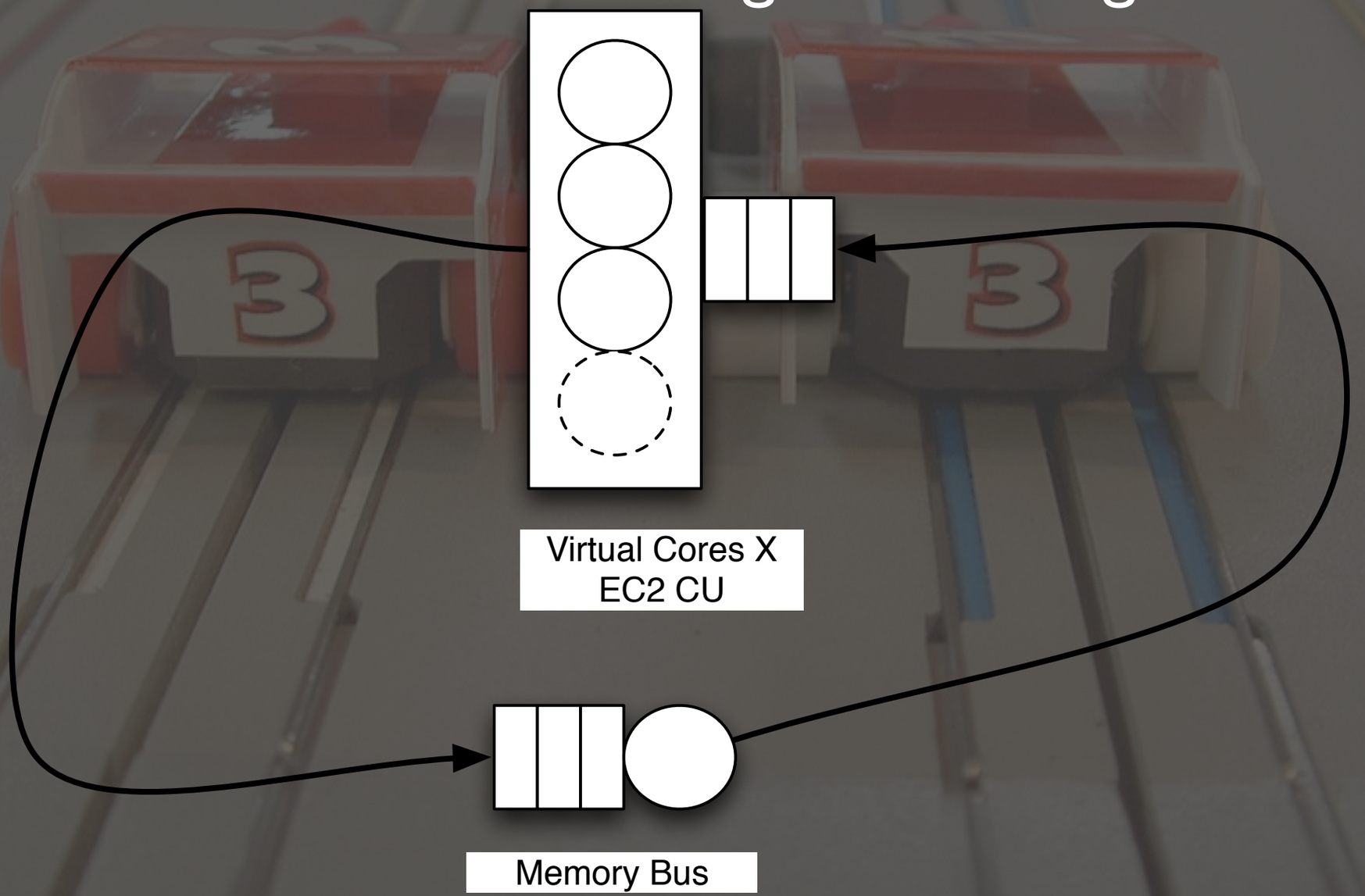
Filas como modelos?

E se trocarmos o disco por um com mais capacidade de IO?



Filas como Modelos

m1.small ? m1.large ? m1.xlarge ?



use pdq;

- Disponível em http://goo.gl/s98wQ
- PDQ é um *queuing circuit solver* por Neil J. Gunther
- Tem um livro inteiro dedicado http://goo.gl/9MA2c



use pdq;

CreateNode()	Define uma fila
CreateOpen()	Define um tráfego de circuito fechado
CreateClosed()	Define um tráfego de circuito aberto
SetDemand()	Define a demanda gerada

use pdq;

Tipos de nós

CEN	Nó de fila
DLY	Nó de latência

use pdq;

Disciplinas

FCFS	First-come first-served
LCFS	Last-come first-served
ISRV	Infinite Server
PSHR	Processor Sharing

use pdq;

- Apache Web Server
- Average Network RTD: 0.00921 seconds
  - Added as a delay center in the circuit
- Average Arrival Rate: 65.142 hits/s
- Average Service time: 0.1159 seconds
- 128 worker threads

use pdq;

```
$workload = "httpd";

$httpMaxClient = 128;

pdq::Init("web server");

$arrivalRate = 65.142;

$serviceTime = 0.1159;

$pdq::streams = \

    pdq::CreateOpen($workload,
    $arrivalRate);
```

pdq::Report();

Metric	Value	Unit
Workload: "httpd"		
Number in system	8.0279	Trans
Mean throughput	65.1420	Trans/Sec
Response time	0.1232	Sec
Stretch factor	1.0626	

pdq::Report();

```
Bounds Analysis:

Max throughput          1104.4003     Trans/Sec

Min response               0.1160     Sec
```

pdq::Report();

- Tamanho médio da requisição: 145 KBytes
  - ~ 1160 Kbits
- @1104 transactions / second:
  - 1,280,640 Kbits /s ~ 1.28 Gbps

Referências

- Performance by Design - Menasce, Dowdy, Almeida - http://amzn.to/mpqfVO
- Capacity Planning for Web Performance: Metrics, Models, and Methods - Daniel Menasce, Virgilio Almeida - http://amzn.to/IOATba
- Capacity Planning for Web Services: Metrics, Models, and Methods - Daniel Menasce, Virgilio Almeida - http://amzn.to/iClpsB

Referências

- Guerrilla Capacity Planning: A Tactical Approach to Planning for Highly Scalable Applications and Services - Neil Gunther - http://amzn.to/kfrfLK
- The Art of Computer Systems Performance Analysis: Techniques for Experimental Design, Measurement, Simulation, and Modeling - R. K. Jain - http://amzn.to/jqud11

Ferramentas

Por que Open Source ?



Por que Open Source ?



Por que Open Source ?

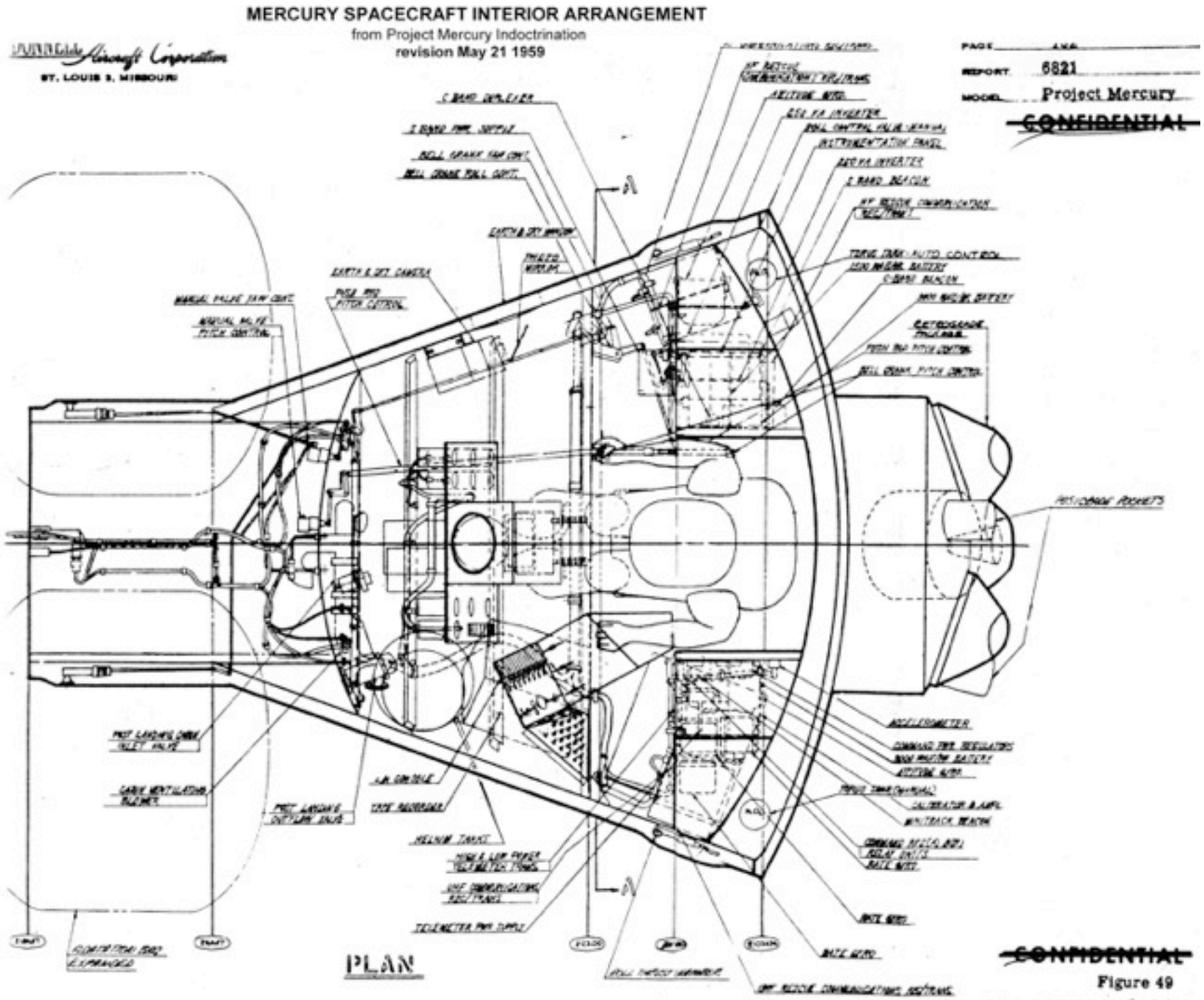


Figure 49

Por que Open Source ?

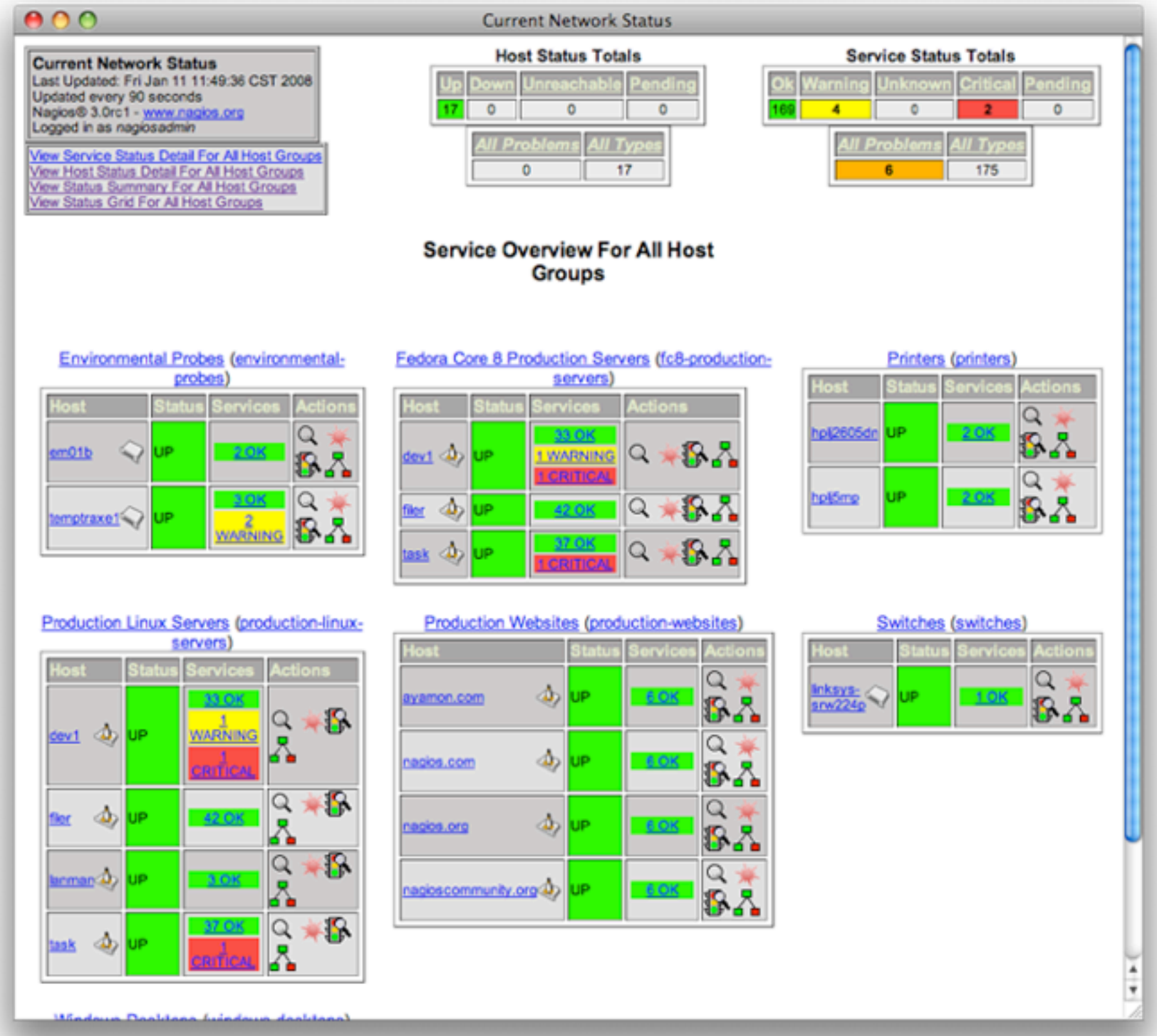


It works !

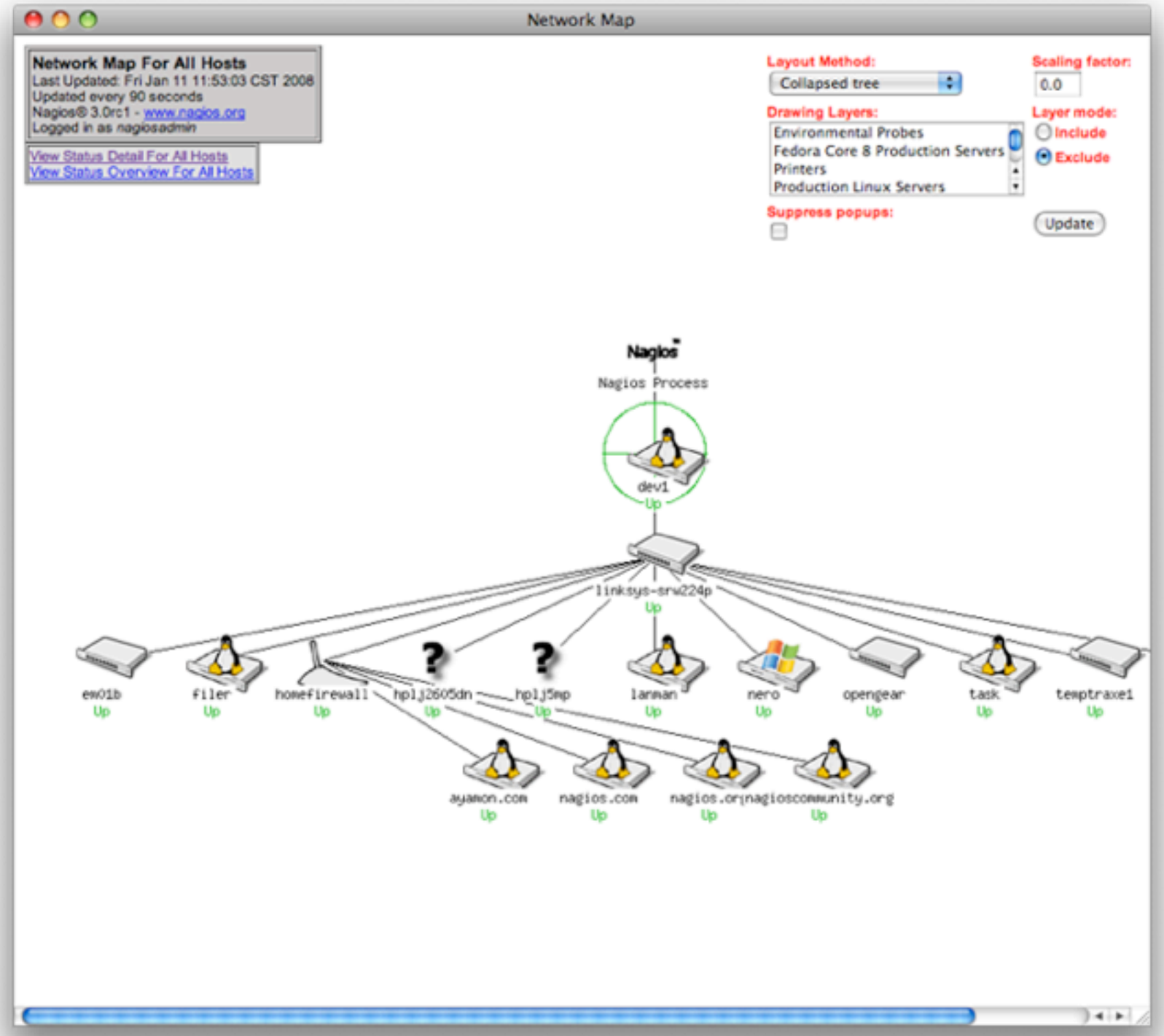
Nagios

- Disponível para praticamente todos os sistemas Unix-like
- Tem se tornado um padrão, com instalações monitorando milhares de servidores e dezenas de milhares de componentes
- Pode monitorar:
  - Linux
  - Windows
  - Serviços HTTP, FTP, SMTP, etc...
  - Equipamentos de rede
- Licenciado sob a GPL
- http://www.nagios.org/

Nagios


Current Network Status
Current Network Status
Last Updated: Fri Jan 11 11:49:36 CST 2008
Updated every 90 seconds
Nagios® 3.0rc1 - www.nagios.org
Logged in as nagiosadmin
View Service Status Detail For All Host Groups
View Host Status Detail For All Host Groups
View Status Summary For All Host Groups
View Status Grid For All Host Groups
Host Status Totals
Up Down Unreachable Pending
17 0 0 0
All Problems All Types
0 17
Service Status Totals
Ok Warning Unknown Critical Pending
169 4 0 2 0
All Problems All Types
6 175
Service Overview For All Host Groups
Environmental Probes (environmental-probes)
Host Status Services Actions
em01b UP 2 OK
temptraxe1 UP 3 OK 2 WARNING
Fedora Core 8 Production Servers (fc8-production-servers)
Host Status Services Actions
dev1 UP 33 OK 1 WARNING 1 CRITICAL
filer UP 42 OK
task UP 37 OK 1 CRITICAL
Printers (printers)
Host Status Services Actions
hpl2605dn UP 2 OK
hpl5mp UP 2 OK
Production Linux Servers (production-linux-servers)
Host Status Services Actions
dev1 UP 33 OK 1 WARNING 1 CRITICAL
filer UP 42 OK
tanman UP 3 OK
task UP 37 OK 1 CRITICAL
Production Websites (production-websites)
Host Status Services Actions
ayamon.com UP 6 OK
nagios.com UP 6 OK
nagios.org UP 6 OK
nagioscommunity.org UP 6 OK
Switches (switches)
Host Status Services Actions
linksys-srw224p UP 1 OK

Nagios


Network Map
Network Map For All Hosts
Last Updated: Fri Jan 11 11:53:03 CST 2008
Updated every 90 seconds
Nagios® 3.0rc1 - www.nagios.org
Logged in as nagiosadmin
View Status Detail For All Hosts
View Status Overview For All Hosts
Layout Method:
Collapsed tree
Scaling factor:
0.0
Drawing Layers:
Environmental Probes
Fedora Core 8 Production Servers
Printers
Production Linux Servers
Layer mode:
Include
Exclude
Suppress popups:
Update
Nagios
Nagios Process
dev1
Up
linksys-srw224p
Up
ew01b
Up
filer
Up
homefirewall
Up
hplj2605dn
Up
hplj5mp
Up
lanman
Up
nero
Up
opengear
Up
task
Up
temptraxe1
Up
ayamon.com
Up
nagios.com
Up
nagios.orgnagioscommunity.org
Up

Nagios

















Zabbix

- Uma alternativa ao Nagios
- Oferece uma gama de monitoração de exceção e de gráficos
- Tem um suporte comercial mais estabelecido
- http://www.zabbix.com/

Zabbix


User defined screens [refreshed every 120 sec] - Mozilla
File Edit View Go Bookmarks Tools Window Help
http://linux.local/~zabbix/screens.php?screenid=1
ZABBIX
View Reports Configuration Login
Latest data | Triggers | Queue | Actions | Alerts | Maps | Graphs | Screens | IT Services |
SCREENS / Zabbix server
Zabbix server
Memory usage (1h history)
Disk load (1h history)
MySQL queries per second (1h history)
Processor loads (1h history)
Network load (1h history)
History growth (1h history)

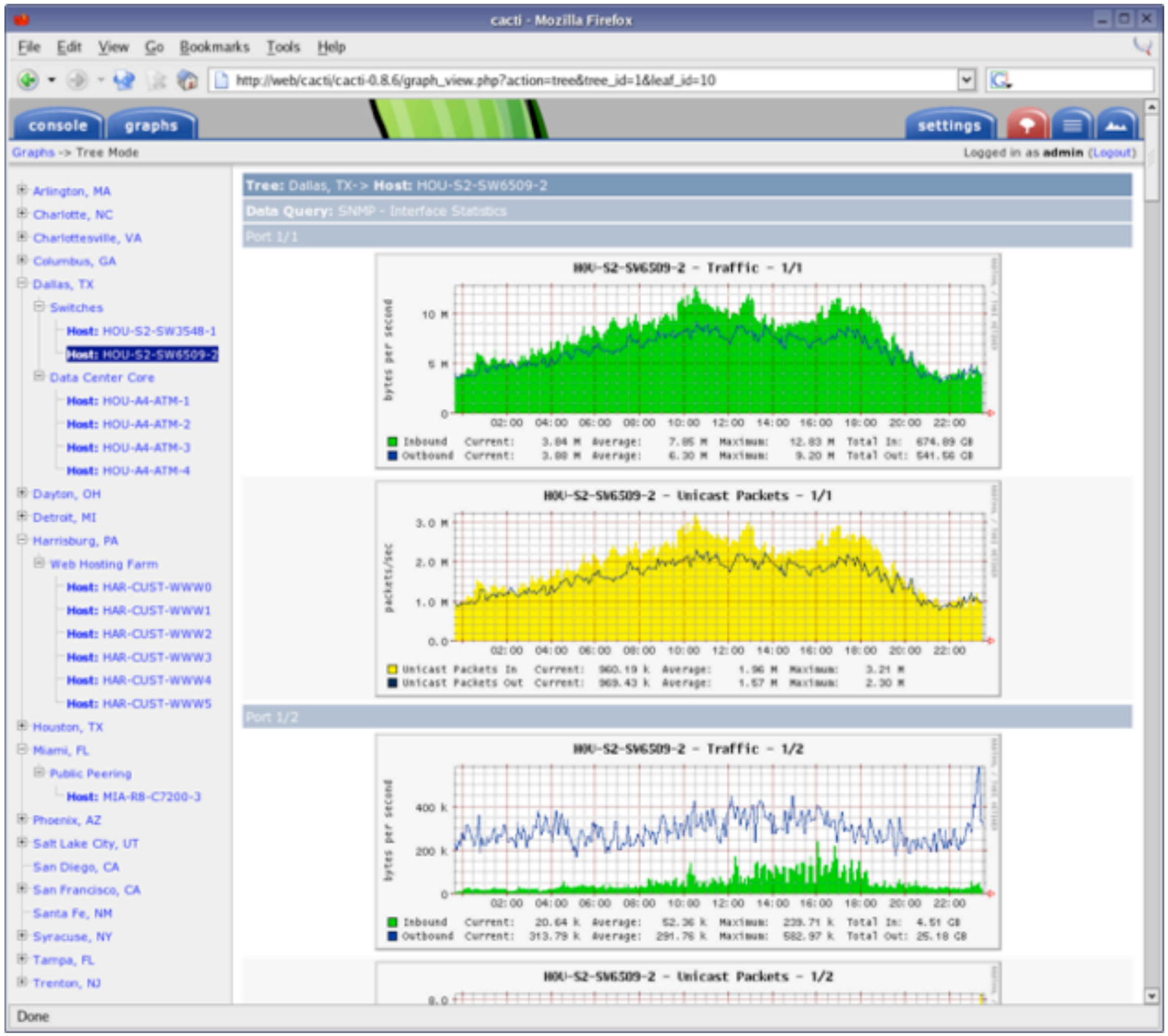
Collectd

- Opera como um daemon que coleta dados de desempenho do sistema
- Pode trabalhar em um regime de proxies
- Payload mínimo para o sistema
- Dezenas de plugins para monitoração de sistemas já disponíveis
- http://collectd.org/

Cacti

- Solução de coleta de dados de desempenho e utilização de recursos
- Geração de gráficos
- Alguns problemas de escalabilidade conhecidos para milhares de hosts
- Usa RRD para persistência dos dados
- http://www.cacti.net/

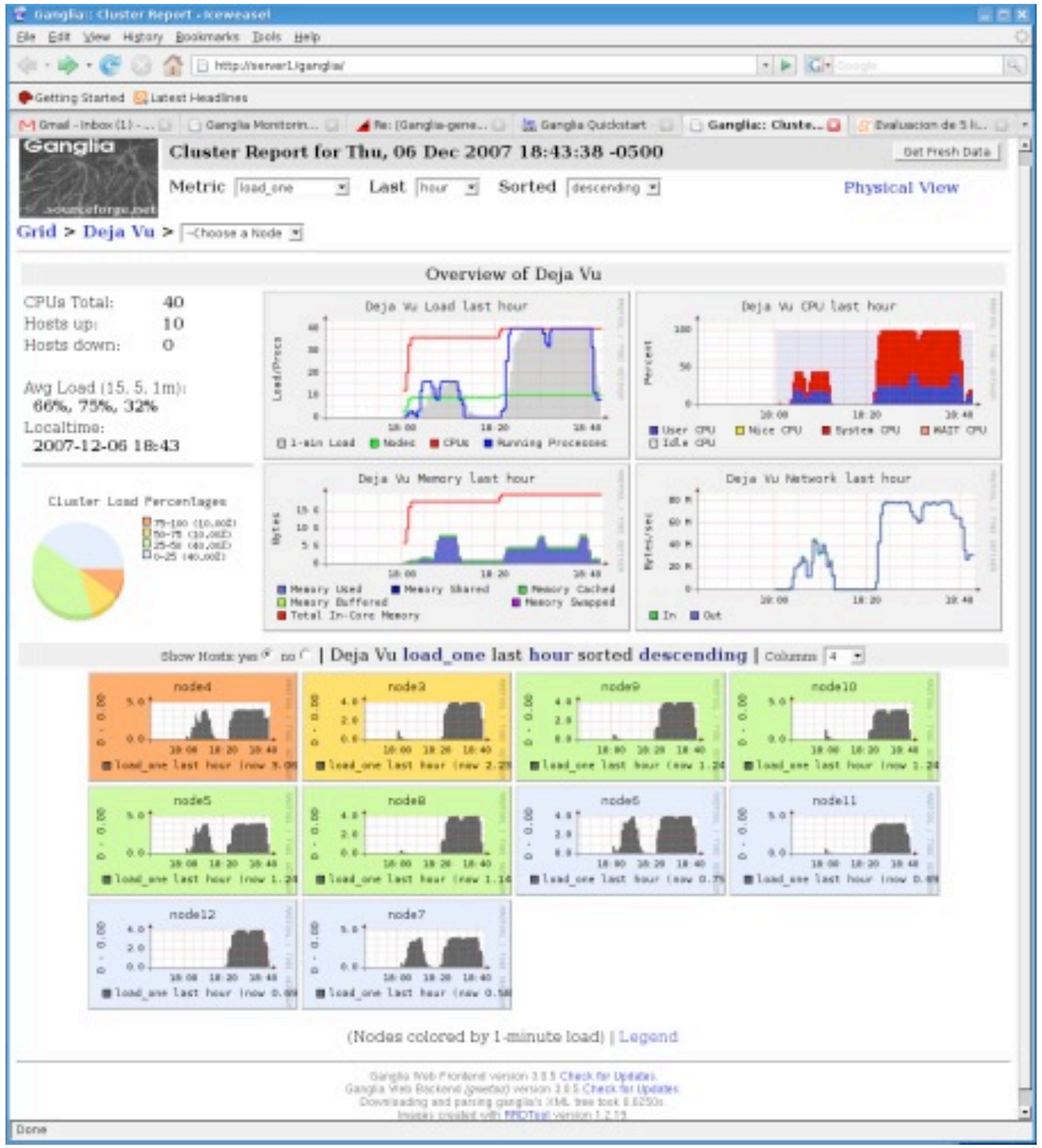
Cacti


cacti - Mozilla Firefox
File Edit View Go Bookmarks Tools Help
http://web/cacti/cacti-0.8.6/graph_view.php?action=tree&tree_id=1&leaf_id=10
console
graphs
settings
Graphs -> Tree Mode
Logged in as admin (Logout)
Arlington, MA
Charlotte, NC
Charlottesville, VA
Columbus, GA
Dallas, TX
Switches
Host: HOU-S2-SW3548-1
Host: HOU-S2-SW6509-2
Data Center Core
Host: HOU-A4-ATM-1
Host: HOU-A4-ATM-2
Host: HOU-A4-ATM-3
Host: HOU-A4-ATM-4
Dayton, OH
Detroit, MI
Harrisburg, PA
Web Hosting Farm
Host: HAR-CUST-WWW0
Host: HAR-CUST-WWW1
Host: HAR-CUST-WWW2
Host: HAR-CUST-WWW3
Host: HAR-CUST-WWW4
Host: HAR-CUST-WWW5
Houston, TX
Miami, FL
Public Peering
Host: MIA-R8-C7200-3
Phoenix, AZ
Salt Lake City, UT
San Diego, CA
San Francisco, CA
Santa Fe, NM
Syracuse, NY
Tampa, FL
Trenton, NJ
Tree: Dallas, TX-> Host: HOU-S2-SW6509-2
Data Query: SNMP - Interface Statistics
Port 1/1
HOU-S2-SW6509-2 - Traffic - 1/1
bytes per second
10 M
5 M
0
02:00 04:00 06:00 08:00 10:00 12:00 14:00 16:00 18:00 20:00 22:00
Inbound Current: 3.84 M Average: 7.05 M Maximum: 12.83 M Total In: 674.89 GB
Outbound Current: 3.88 M Average: 6.30 M Maximum: 9.20 M Total Out: 541.56 GB
HOU-S2-SW6509-2 - Unicast Packets - 1/1
packets/sec
3.0 M
2.0 M
1.0 M
0.0
02:00 04:00 06:00 08:00 10:00 12:00 14:00 16:00 18:00 20:00 22:00
Unicast Packets In Current: 960.19 k Average: 1.96 M Maximum: 3.21 M
Unicast Packets Out Current: 969.43 k Average: 1.57 M Maximum: 2.30 M
Port 1/2
HOU-S2-SW6509-2 - Traffic - 1/2
bytes per second
400 k
200 k
0
02:00 04:00 06:00 08:00 10:00 12:00 14:00 16:00 18:00 20:00 22:00
Inbound Current: 20.64 k Average: 52.36 k Maximum: 239.71 k Total In: 4.51 GB
Outbound Current: 313.79 k Average: 291.76 k Maximum: 582.97 k Total Out: 25.18 GB
HOU-S2-SW6509-2 - Unicast Packets - 1/2
0.0
Done

Ganglia

- Projeto que nasceu orientado para monitoração de clusters de HPC
- Oferece opções de agregação de dados para clusters facilitando a visualização de recursos o sistema todo
- Tem uma escalabilidade maior do que o Cacti porém com uma interface mais pobre

Ganglia


Ganglia :: Cluster Report - Iceweasel
Cluster Report for Thu, 06 Dec 2007 18:43:38 -0500
Get Fresh Data
Metric load_one
Last hour
Sorted descending
Physical View
Grid > Deja Vu > -Choose a Node
Overview of Deja Vu
CPUs Total: 40
Hosts up: 10
Hosts down: 0
Avg Load (15, 5, 1m):
66%, 75%, 32%
Localtime:
2007-12-06 18:43
Deja Vu Load last hour
Deja Vu CPU last hour
Cluster Load Percentages
Deja Vu Memory last hour
Deja Vu Network last hour
Show Hosts: yes no | Deja Vu load_one last hour sorted descending | Columns 4
node4
node3
node9
node10
node5
node8
node6
node11
node12
node7
(Nodes colored by 1-minute load) | Legend

Ganglia


THE UNIVERSITY OF CALIFORNIA BERKELEY
LET THERE BE LIGHT
1868


twitter


LOCKHEED MARTIN


flickr™


BOEING®


NASA


DOW®


CERN


Bank of America

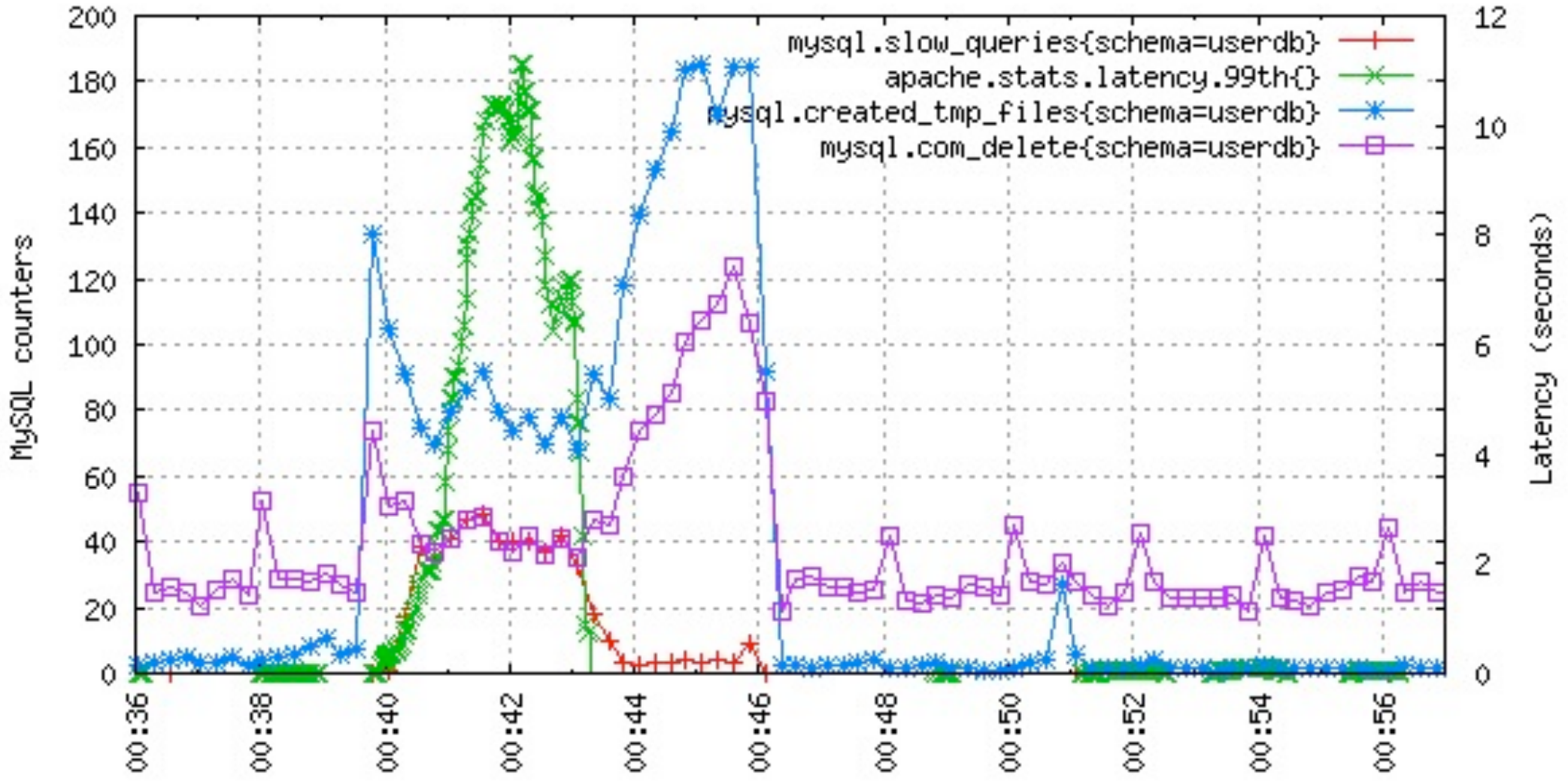
SYSSTAT

- Pacote *standard* em distribuições Unix atuais
- Geralmente conhecido por apenas um dos seus componentes (sar)
- Componentes:
  - sar/sadc/sadf
  - iostat / nfsiostat / cifsiostat
  - mpstat
  - pidstat

OpenTSDB

- Usa o Hadoop (Hbase) para persistência dos dados
- Se propõe a monitorar milhares de hosts e aplicações sem necessariamente perder precisão de dados históricos
- http://opentsdb.net/

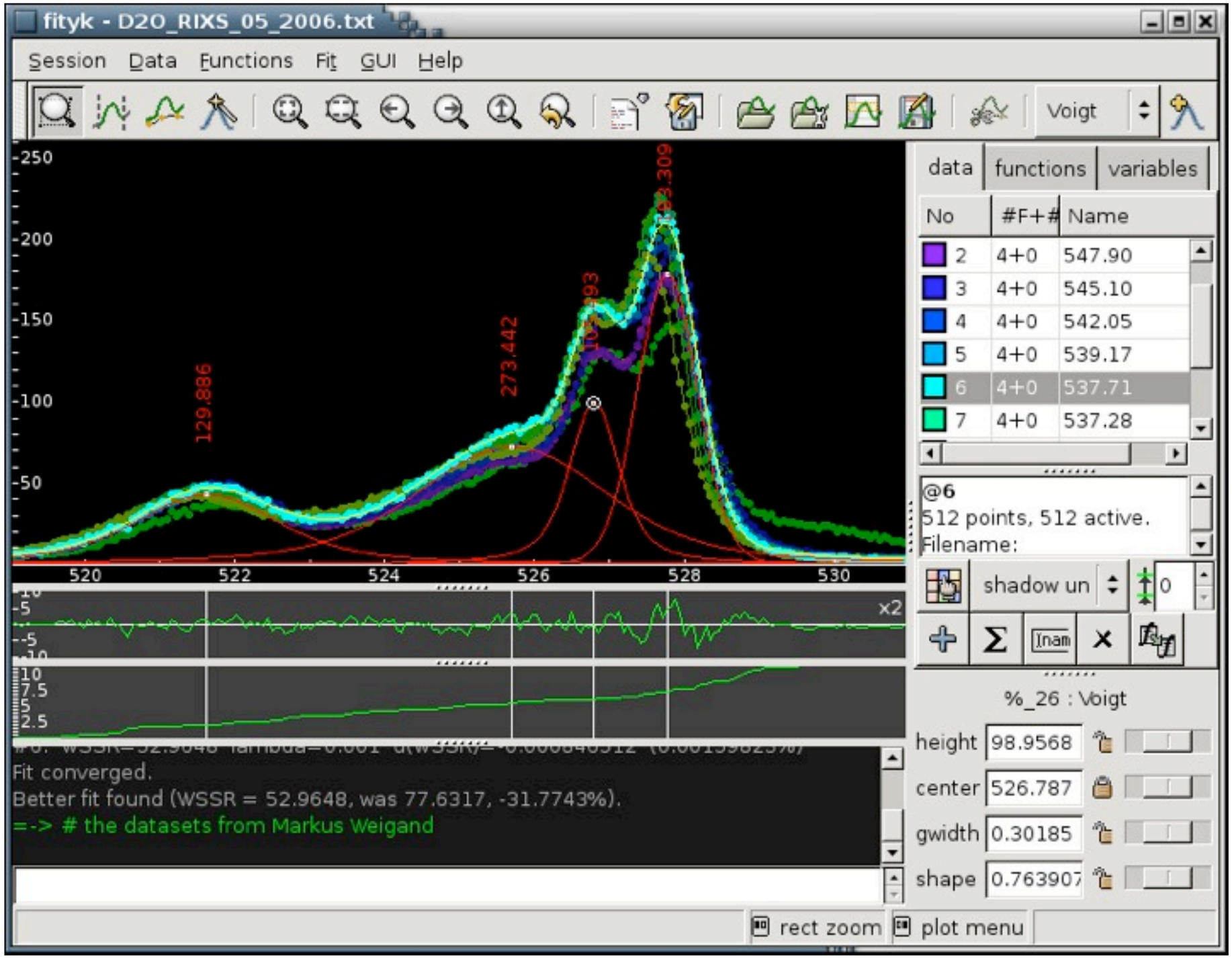
OpenTSDB


mysql.slow_queries{schema=userdb}
apache.stats.latency.99th{}
mysql.created_tmp_files{schema=userdb}
mysql.com_delete{schema=userdb}
MySQL counters
Latency (seconds)
200
180
160
140
120
100
80
60
40
20
0
12
10
8
6
4
2
0
00:36
00:38
00:40
00:42
00:44
00:46
00:48
00:50
00:52
00:54
00:56

Fityk

- *Curve Fitting*
- Ferramenta para geração de gráfico e análise de tendência
- Versões para Windows, Linux e Mac
- Permite a automação da coleta dos dados via linha de comando
- http://fityk.nieto.pl/

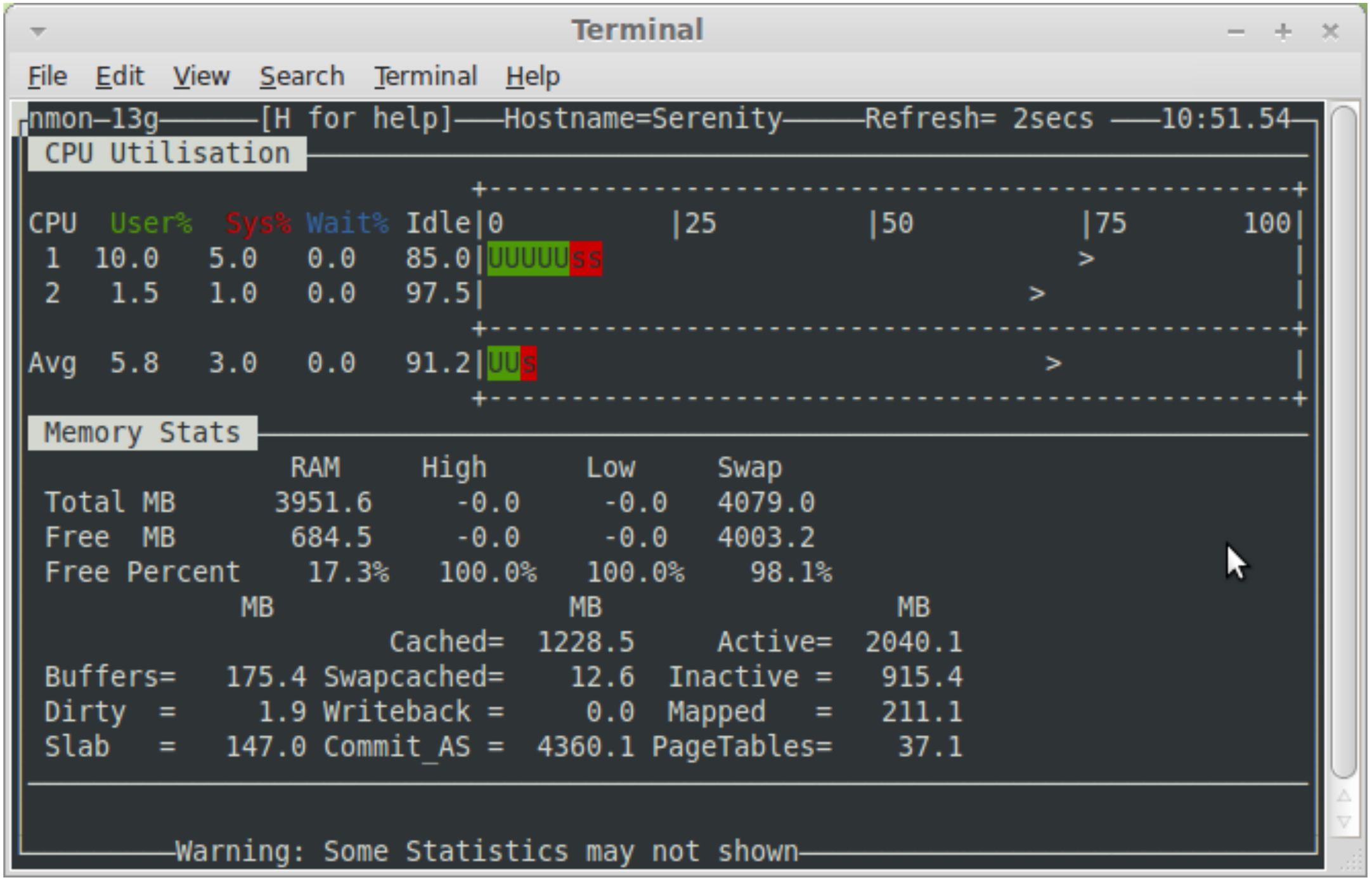
Fityk


fityk - D20_RIXS_05_2006.txt
Session Data Functions Fit GUI Help
Voigt
data functions variables
No #F+# Name
2 4+0 547.90
3 4+0 545.10
4 4+0 542.05
5 4+0 539.17
6 4+0 537.71
7 4+0 537.28
@6
512 points, 512 active.
Filename:
shadow un
0
%_26 : Voigt
height 98.9568
center 526.787
gwidth 0.30185
shape 0.763907
129.886
273.442
309
193
-250
-200
-150
-100
-50
520
522
524
526
528
530
x2
10
7.5
5
2.5
Fit converged.
Better fit found (WSSR = 52.9648, was 77.6317, -31.7743%).
=-> # the datasets from Markus Weigand
rect zoom
plot menu

NMon

- Ferramenta para acompanhamento em tempo real dos contadores de desempenho do sistema
- Permite exportar os dados para um CSV para análise posterior ou geração de gráficos
- http://nmon.sourceforge.net/pmwiki.php

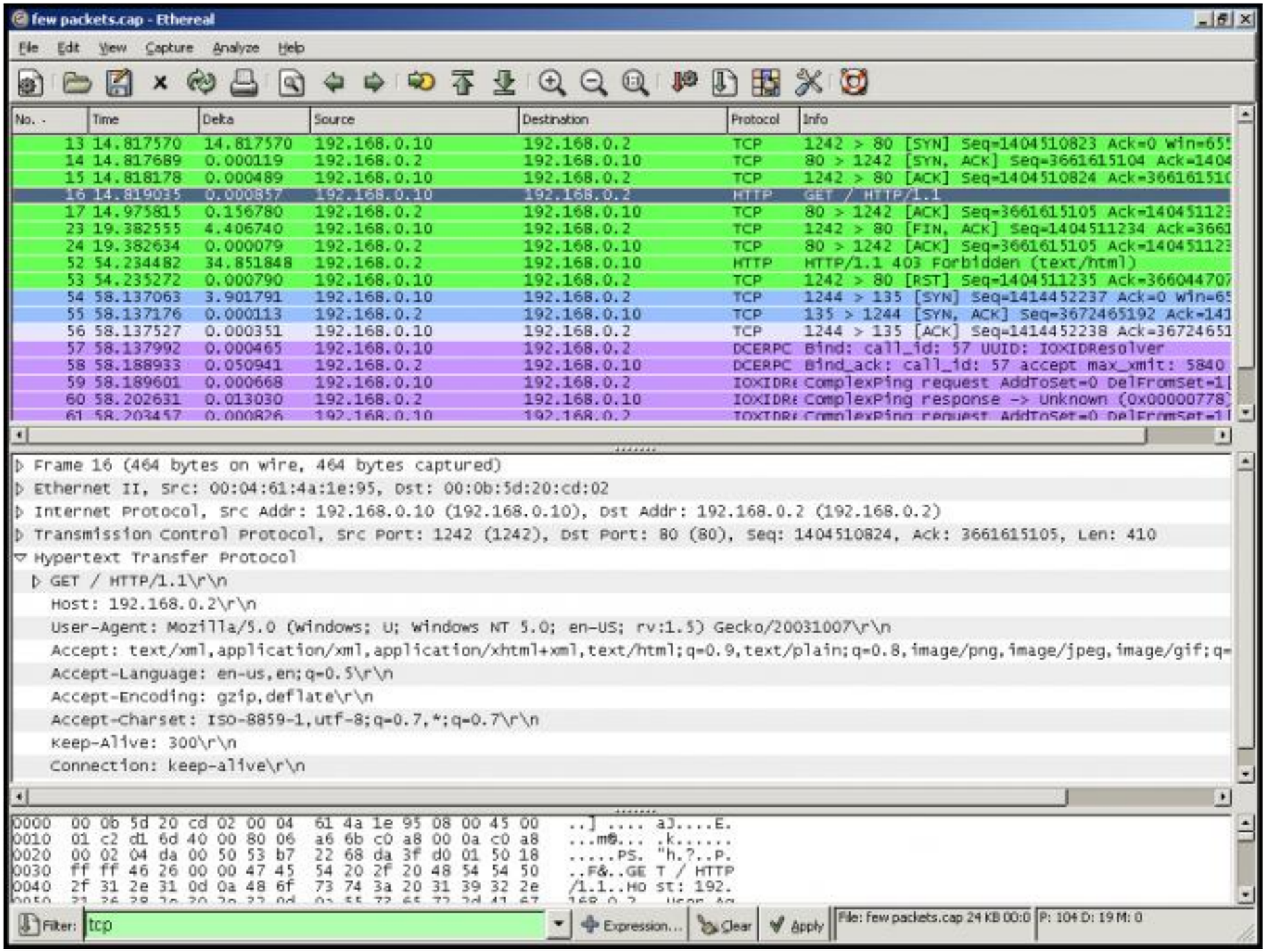
NMon


Terminal
File Edit View Search Terminal Help
nmon-13g [H for help] Hostname=Serenity Refresh= 2secs 10:51.54
CPU Utilisation
CPU User% Sys% Wait% Idle|0 |25 |50 |75 100|
1 10.0 5.0 0.0 85.0|UUUUUss
2 1.5 1.0 0.0 97.5|
Avg 5.8 3.0 0.0 91.2|UUs
Memory Stats
RAM High Low Swap
Total MB 3951.6 -0.0 -0.0 4079.0
Free MB 684.5 -0.0 -0.0 4003.2
Free Percent 17.3% 100.0% 100.0% 98.1%
MB MB MB
Cached= 1228.5 Active= 2040.1
Buffers= 175.4 Swapcached= 12.6 Inactive = 915.4
Dirty = 1.9 Writeback = 0.0 Mapped = 211.1
Slab = 147.0 Commit_AS = 4360.1 PageTables= 37.1
Warning: Some Statistics may not shown

Wireshark

- Conhecido anteriormente por *Ethereal*
- Permite uma análise detalhada do tráfego de rede
- Pode analisar arquivos com o tráfego capturado pelo *tcpdump*
- http://www.wireshark.org/

Wireshark

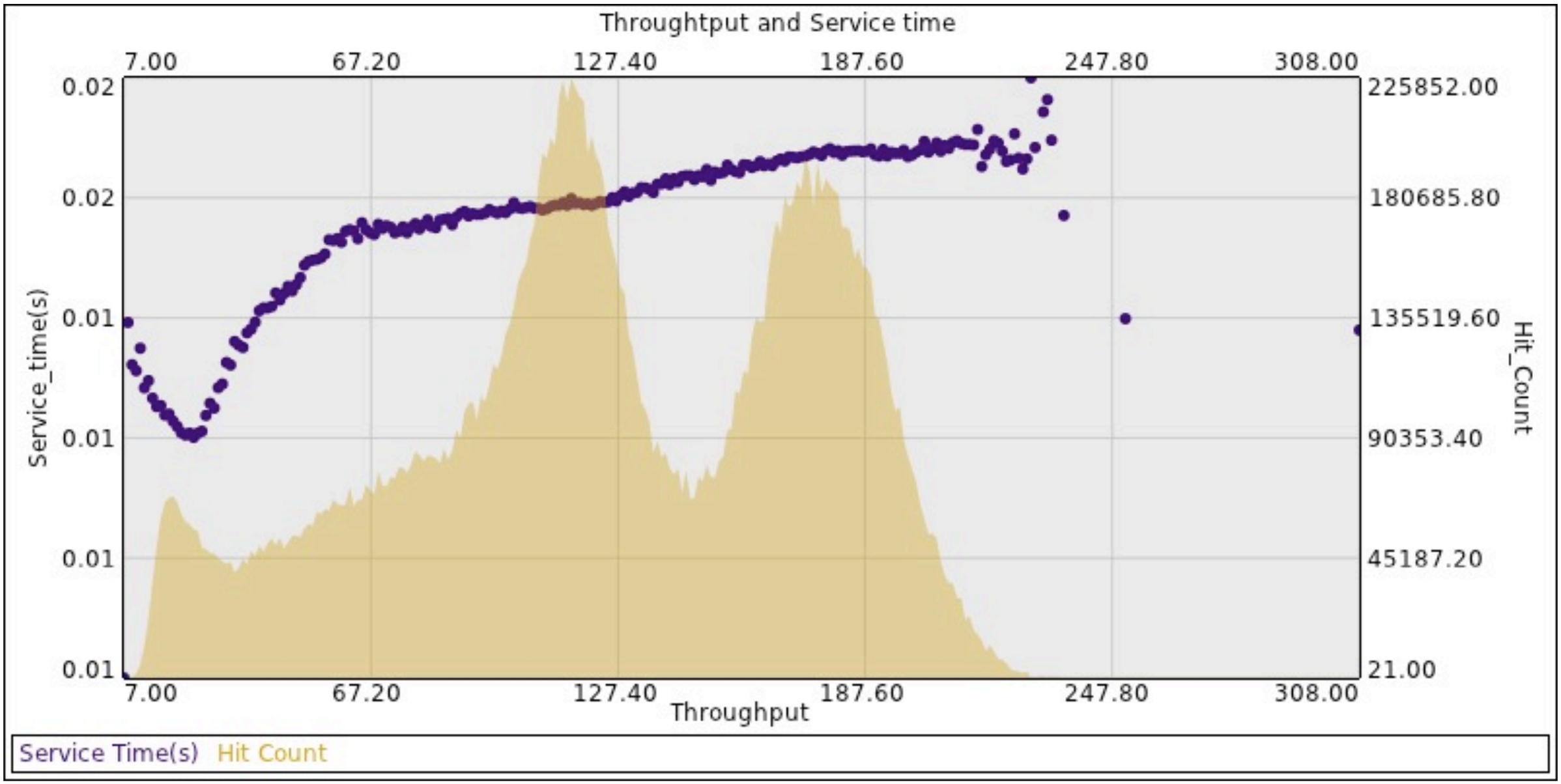

few packets.cap - Ethereal
File Edit View Capture Analyze Help
No. - Time Delta Source Destination Protocol Info
13 14.817570 14.817570 192.168.0.10 192.168.0.2 TCP 1242 > 80 [SYN] Seq=1404510823 Ack=0 Win=65
14 14.817689 0.000119 192.168.0.2 192.168.0.10 TCP 80 > 1242 [SYN, ACK] Seq=3661615104 Ack=1404
15 14.818178 0.000489 192.168.0.10 192.168.0.2 TCP 1242 > 80 [ACK] Seq=1404510824 Ack=366161510
16 14.819035 0.000857 192.168.0.10 192.168.0.2 HTTP GET / HTTP/1.1
17 14.975815 0.156780 192.168.0.2 192.168.0.10 TCP 80 > 1242 [ACK] Seq=3661615105 Ack=140451123
23 19.382555 4.406740 192.168.0.10 192.168.0.2 TCP 1242 > 80 [FIN, ACK] Seq=1404511234 Ack=3661
24 19.382634 0.000079 192.168.0.2 192.168.0.10 TCP 80 > 1242 [ACK] Seq=3661615105 Ack=140451123
52 54.234482 34.851848 192.168.0.2 192.168.0.10 HTTP HTTP/1.1 403 Forbidden (text/html)
53 54.235272 0.000790 192.168.0.10 192.168.0.2 TCP 1242 > 80 [RST] Seq=1404511235 Ack=366044707
54 58.137063 3.901791 192.168.0.10 192.168.0.2 TCP 1244 > 135 [SYN] Seq=1414452237 Ack=0 Win=65
55 58.137176 0.000113 192.168.0.2 192.168.0.10 TCP 135 > 1244 [SYN, ACK] Seq=3672465192 Ack=141
56 58.137527 0.000351 192.168.0.10 192.168.0.2 TCP 1244 > 135 [ACK] Seq=1414452238 Ack=36724651
57 58.137992 0.000465 192.168.0.10 192.168.0.2 DCERPC Bind: call_id: 57 UUID: IOXIDResolver
58 58.188933 0.050941 192.168.0.2 192.168.0.10 DCERPC Bind_ack: call_id: 57 accept max_xmit: 5840
59 58.189601 0.000668 192.168.0.10 192.168.0.2 IOXIDR ComplexPing request AddToSet=0 DelFromSet=1
60 58.202631 0.013030 192.168.0.2 192.168.0.10 IOXIDR ComplexPing response -> Unknown (0x00000778)
Frame 16 (464 bytes on wire, 464 bytes captured)
Ethernet II, Src: 00:04:61:4a:1e:95, Dst: 00:0b:5d:20:cd:02
Internet Protocol, Src Addr: 192.168.0.10 (192.168.0.10), Dst Addr: 192.168.0.2 (192.168.0.2)
Transmission Control Protocol, Src Port: 1242 (1242), Dst Port: 80 (80), Seq: 1404510824, Ack: 3661615105, Len: 410
Hypertext Transfer Protocol
GET / HTTP/1.1\r\n
Host: 192.168.0.2\r\n
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.0; en-US; rv:1.5) Gecko/20031007\r\n
Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,image/png,image/jpeg,image/gif;q=
Accept-Language: en-us,en;q=0.5\r\n
Accept-Encoding: gzip,deflate\r\n
Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7\r\n
Keep-Alive: 300\r\n
Connection: keep-alive\r\n
0000 00 0b 5d 20 cd 02 00 04 61 4a 1e 95 08 00 45 00 ..] .... aJ....E.
0010 01 c2 d1 6d 40 00 80 06 a6 6b c0 a8 00 0a c0 a8 ...m@... .k......
0020 00 02 04 da 00 50 53 b7 22 68 da 3f d0 01 50 18 .....PS. "h.?..P.
0030 ff ff 46 26 00 00 47 45 54 20 2f 20 48 54 54 50 ..F&..GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 31 39 32 2e /1.1..Ho st: 192.
Filter: tcp
Expression... Clear Apply
File: few packets.cap 24 KB 00:0
P: 104 D: 19 M: 0

Fork me on GitHub

HPA

- HTTP Performance Analyzer
- Gera um gráfico de desempenho de servidores HTTP
- Precisa que o último campo do log seja o *service time* da transação
- https://github.com/camposr/HTTP-Performance-Analyzer

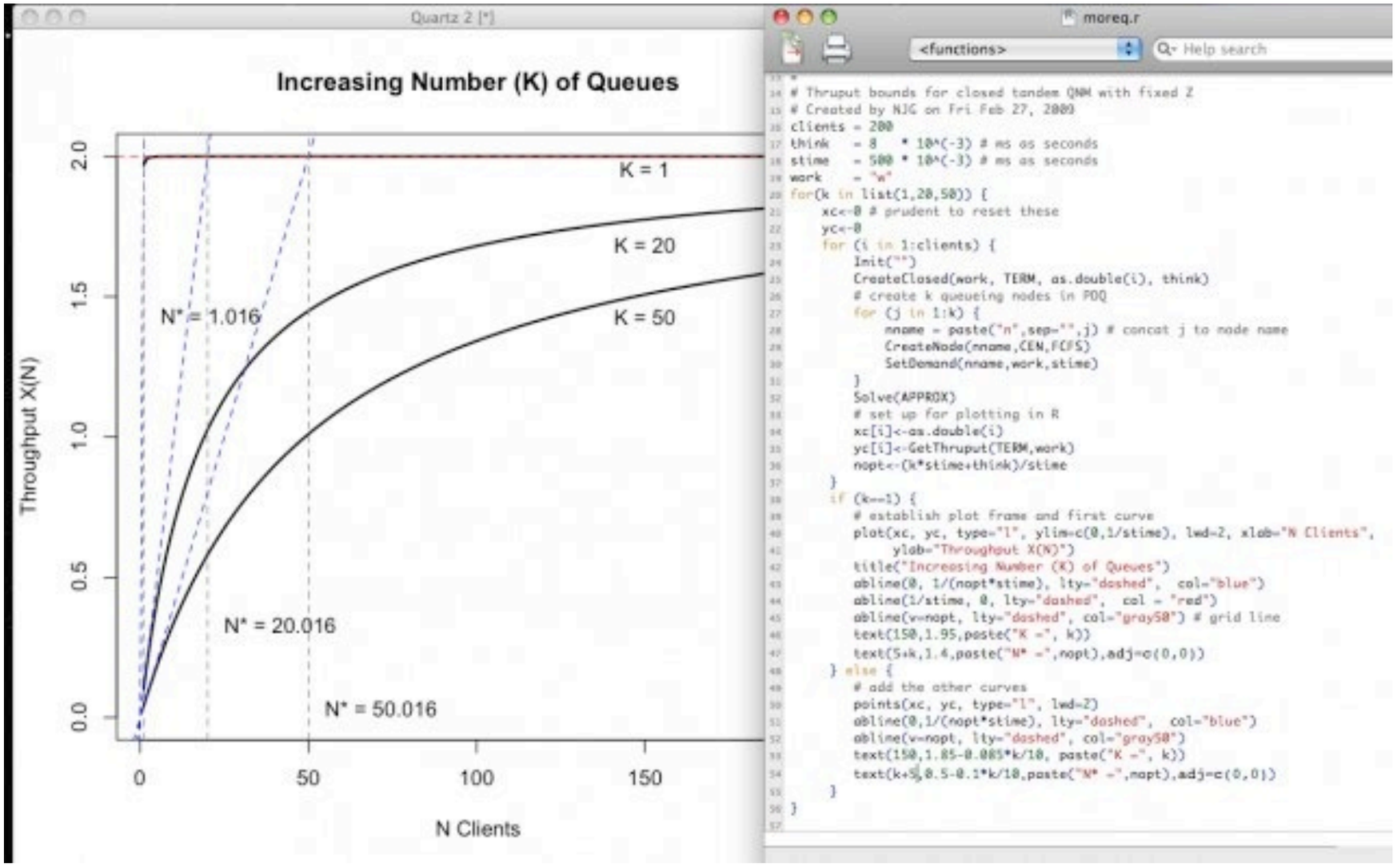
HPA


Throughtput and Service time
7.00
67.20
127.40
187.60
247.80
308.00
0.02
0.02
0.01
0.01
0.01
225852.00
180685.80
135519.60
90353.40
45187.20
21.00
Service_time(s)
Hit_Count
Throughput
Service Time(s)
Hit Count

PDQ

- Ferramenta para resolução de modelos de fila
- Permite a criação de *what-if scenarios*
- *Bindings* para diversas linguagens e para o R
- http://sourceforge.net/projects/pdq-qnm-pkg/

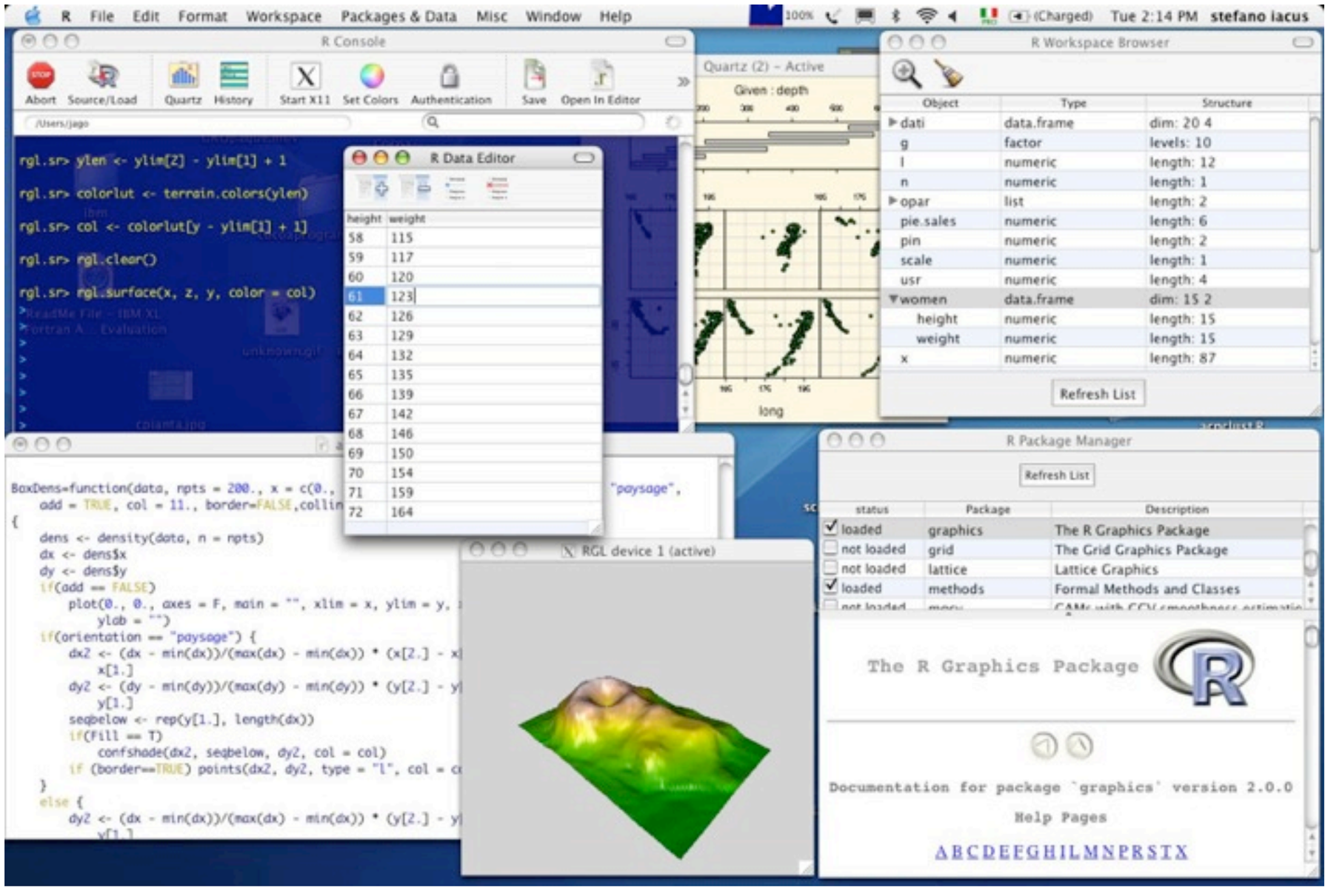
PDQ


Quartz 2 [*]
Increasing Number (K) of Queues
K = 1
K = 20
K = 50
N* = 1.016
N* = 20.016
N* = 50.016
Throughput X(N)
N Clients
0.0
0.5
1.0
1.5
2.0
0
50
100
150
moreq.r
<functions>
Help search
Thruput bounds for closed tandem QNM with fixed Z
Created by NJG on Fri Feb 27, 2003
clients = 200
think = 8 * 10^(-3) # ms as seconds
stime = 500 * 10^(-3) # ms as seconds
work = "w"
for(k in list(1,20,50)) {
xc<-0 # prudent to reset these
yc<-0
for (i in 1:clients) {
Init("")
CreateClosed(work, TERM, as.double(i), think)
create k queueing nodes in PDQ
for (j in 1:k) {
nname = paste("n",sep="",j) # concat j to node name
CreateNode(nname,CEN,FCFS)
SetDemand(nname,work,stime)
}
Solve(APPROX)
set up for plotting in R
xc[i]<-as.double(i)
yc[i]<-GetThruput(TERM,work)
nopt<-(k*stime+think)/stime
}
if (k==1) {
establish plot frame and first curve
plot(xc, yc, type="l", ylim=c(0,1/stime), lwd=2, xlab="N Clients",
ylab="Throughput X(N)")
title("Increasing Number (K) of Queues")
abline(0, 1/(nopt*stime), lty="dashed", col="blue")
abline(1/stime, 0, lty="dashed", col = "red")
abline(v=nopt, lty="dashed", col="gray50") # grid line
text(150,1.95,paste("K =", k))
text(5+k,1.4,paste("N* =",nopt),adj=c(0,0))
} else {
add the other curves
points(xc, yc, type="l", lwd=2)
abline(0,1/(nopt*stime), lty="dashed", col="blue")
abline(v=nopt, lty="dashed", col="gray50")
text(150,1.85-0.085*k/10, paste("K =", k))
text(k+5,0.5-0.1*k/10,paste("N* =",nopt),adj=c(0,0))
}
}

R

- The R Project for Statistical Computing
- Um pacote completo para análise estatística e geração de gráficos
- Disponível para Linux, Windows e Mac
- Centenas de módulos estatísticos disponíveis via CRAN
- http://www.r-project.org/

R


R File Edit Format Workspace Packages & Data Misc Window Help
Tue 2:14 PM stefano iacus
R Console
Abort Source/Load Quartz History Start X11 Set Colors Authentication Save Open In Editor
rgl.sr> ylen <- ylim[2] - ylim[1] + 1
rgl.sr> colorlut <- terrain.colors(ylen)
rgl.sr> col <- colorlut[y - ylim[1] + 1]
rgl.sr> rgl.clear()
rgl.sr> rgl.surface(x, z, y, color = col)
R Data Editor
height weight
58 115
59 117
60 120
61 123
62 126
63 129
64 132
65 135
66 139
67 142
68 146
69 150
70 154
71 159
72 164
Quartz (2) - Active
Given : depth
long
R Workspace Browser
Object Type Structure
dati data.frame dim: 20 4
g factor levels: 10
l numeric length: 12
n numeric length: 1
opar list length: 2
pie.sales numeric length: 6
pin numeric length: 2
scale numeric length: 1
usr numeric length: 4
women data.frame dim: 15 2
height numeric length: 15
weight numeric length: 15
x numeric length: 87
Refresh List
R Package Manager
Refresh List
status Package Description
loaded graphics The R Graphics Package
not loaded grid The Grid Graphics Package
not loaded lattice Lattice Graphics
loaded methods Formal Methods and Classes
RGL device 1 (active)
The R Graphics Package
Documentation for package `graphics' version 2.0.0
Help Pages
A B C D E F G H I L M N P R S T X

Lista completa com endereços

http://bit.ly/caplist

Perguntas ?